

Adaptive prediction for functional time series

Hassan Maissoro^{1,2}, Valentin Patilea¹, and Myriam Vimond¹

¹Univ Rennes, Ensai, CNRS, CREST (Center for Research in Economics and Statistics)
- UMR 9194, F-35000 Rennes, France

²Datastorm, F-91120 Palaiseau - France

August 24, 2026

Abstract

Key words: TODO

Contents

1	Introduction	2
2	Functional time series prediction	2
2.1	Model with measurement errors	3
2.2	Notation	3
2.3	Regularized linear predictor	4
2.4	Choice of α	6
2.5	Prediction interval	7
3	Numerical study	8
3.1	Simulation setting	8
3.2	Implementation details	9
3.3	Autocovariance function estimation	11
3.4	Covariance function estimation	11
3.5	Adaptive curve prediction	13
3.5.1	Empirical autoregressive effect	13
3.5.2	Monte Carlo study	14
3.6	Real data application	17
A	Proofs associated to the Regularized linear predictor	20
B	Uniform consistent estimators	24

1 Introduction

Some facts :

- When the curves are fully observed, a linear predictor have a closed and intuitive form,

$$\mathfrak{X}_{N+1} = \mu + \psi(X_N - \mu)$$

where ψ is a linear bounded operator. This predictor is a BLUP is if it is solution of the moment equation $C_1 = \psi C_0$. However C_0 is compact and its eigenvalues accumulates at 0. Solving the moment equation is an ill-posed problem : if a solution exist, this solution may be not unique.

- [Kühnert et al. \(2026\)](#) consider the estimation of ψ in FAR(1) model :

$$\widehat{\psi}(\cdot, K, \alpha_N) = \widehat{C}_1(\widehat{C}_0 + \alpha_N \text{Id})^{-1} \prod_{\widehat{\phi}_1}^{\widehat{\phi}_K}$$

where $K = K_N \rightarrow \infty$, $\alpha = \alpha_N \rightarrow 0$, $\prod_{\phi_1}^{\phi_K}$ is the orthogonal projection on the K eigenfunctions $\{\phi_i\}$ of C_0 associated to the K largest eigenvalues. Their assumptions are :

- C_0 is injective (Proposition 3.1) and its eigenvalues admit a polynomial decay of degree a ;
- the eigenvalues $\{\beta_k\}$ of ψ satisfy $\sum_k k^{2\beta} \beta_k^2 < \infty$.

Corollary 5.1 ensures that K and α can be chosen so that,

$$\|\widehat{\psi} - \psi\|_{HS} = \mathcal{O}_{\mathbb{P}}(N^{-\beta/(5+4a+2\beta)})$$

There is a double regularization (Tikhonov and projection onto the first K eigenfunctions of C_0).

- [Zhao \(2026\)](#) consider the estimation of ψ in FAR(1) model :

$$\widehat{\psi}(\cdot, \alpha_N) = \widehat{C}_1(\widehat{C}_0 + \alpha_N \text{Id})^{-1}$$

where $\alpha = \alpha_N \rightarrow 0$. Their assumptions are :

- C_0 is injective and its eigenvalues admit a polynomial decay of degree a ;
- $\psi = FC_0^\beta$ with $\beta > 0$ and $\|F\|_{HS} \leq \rho$.

$\alpha_N \sim N^{-1/(2 \min(\beta, 1) + 2)}$ would be the optimal rate in Tikhonov statistique regularization (Cavalier, Bissantz–Hohage–Munk, Engl–Hanke–Neubauer). α_N is selected by cross validation. Theorem 1 ensures that α can be chosen so that,

$$\|\widehat{\psi} - \psi\|_{HS} = \mathcal{O}_{\mathbb{P}}(N^{-\min(\beta, 1)/(2+2 \min(\beta, 1))})$$

There is only one regularization (Tikhonov).

2 Functional time series prediction

We model functional data as noisy sampled points from a collection of trajectories that are assumed to be a strictly stationary functional time series $\{X_n\} = \{X_n(u), u \in I, n \in \mathbb{Z}\}$. Typically, I is a bounded domain on the real line on which the random functions are defined, and without loss of generality we consider $I = (0, 1]$. The processes X_n belong to the Hilbert space $\mathcal{H} = \mathbb{L}^2(I)$ of real-valued square integrable functions defined on I . Moreover, almost surely, the paths X_n are assumed to belong to the Banach space $\mathcal{C} = \mathcal{C}(I)$ of continuous functions, equipped with the sup-norm $\|\cdot\|_\infty$.

Throughout, we assume there exist $p \geq 2$ such that $\nu_p(\|X_0\|_\infty) < \infty$, where $\nu_p(Z) = (\mathbb{E}[|Z|^p])^{1/p} < \infty$ for real valued random variable Z . We denote respectively as $\mathbb{L}_{\mathcal{H}}^p$ and $\mathbb{L}_{\mathcal{C}}^p$ the spaces of $\mathcal{H}$ -valued and $\mathcal{C}$ -valued random functions X such that $\nu_p(\|X\|_{\mathcal{H}}) < \infty$ and $\nu_p(\|X\|_\infty) < \infty$, respectively. Then the mean of X_n is well defined, belongs to $\mathcal{H}$ and is independent of n by stationarity,

$$\mu(t) = \mathbb{E}X_n(t), \quad t \in I.$$

The autocovariance function at lag $\ell \geq 0$ is also defined by,

$$c_\ell(s, t) = \mathbb{E}[\{X_n(s) - \mu(s)\}\{X_{n+\ell}(t) - \mu(t)\}], \quad \forall s, t \in I.$$

2.1 Model with measurement errors

The model of the observations incorporates uncorrelated measurement errors with mean 0 and bounded variance. Let $Y_{n,i}$ be the i th observation of the random function $X_n(\cdot)$ made at the design point $T_{n,i} \in I$. The model is,

$$Y_{n,i} = X_n(T_{n,i}) + \sigma(T_{n,i})\varepsilon_{n,i}, \quad 1 \leq n \leq N, \quad 1 \leq i \leq M_n, \quad (1)$$

where $\varepsilon_{n,i}$ are independent copies of a centred variable ε with unit variance, and $\sigma(t)$ is an unknown bounded function that accounts for possible heteroscedastic measurement errors. The following hypotheses formalise the data generation process described in (1).

- (H1) The series $\{X_n\}$ is a (strictly) stationary $\mathcal{H}$ -valued series.
- (H2) The $M_1, \dots, M_N$ are random draws of an integer-valued variable¹ $M \geq 2$, with the expectation λ .
- (H3) The design points are : either, for all n it holds $M_n = \lambda$ and $T_{n,i} = i/\lambda$, $1 \leq i \leq n$ (common design case); or, all the design $M_1 + \dots + M_N$ points are independent copies of a variable T which admits a positive density with derives up to order r for some integer $r > 1$ (independent design case). Moreover, $g_{\min} = \inf_{t \in I} g(t) > 0$ and the derivative $g^{(r)}$ of order r is bounded.
- (H4) The $\varepsilon_{n,i}$ are independent copies of a centered sub-gaussian error variable ε with unit variance, and $\sigma^2(\cdot)$ is a Lipschitz continuous function.
- (H5) The series $\{X_n\}$ and the copies of M , T and ε are mutually independent.

2.2 Notation

In the following, the subscript M, T for the expectation, variance, and covariance operators denotes the latter operator conditional on M and T . For example, $\mathbb{E}_{M,T}(\cdot)$ denotes the conditional expectation given M_n and $\{T_{n,1}, \dots, T_{n,M_n}\}$ for $n \geq 1$:

$$\mathbb{E}_{M,T}(\cdot) = \mathbb{E}(\cdot \mid M_n, \{T_{n,i}, 1 \leq i \leq M_n\}, 1 \leq n \leq N).$$

Similarly, $\text{Var}_{M,T}(\cdot)$ and $\text{Cov}_{M,T}(\cdot, \cdot)$ denote the associated conditional variance and covariance operators. Let Q_n be a weighted empirical measure with the mass concentrated on $\{T_{n,i}, i = 1 \dots M_n\}$, that is

$$Q_n = \sum_{i=1}^{M_n} \varrho_{n,i} \delta_{T_{n,i}}, \quad \text{with} \quad \varrho_{n,i} > 0, \quad \sum_{i=1}^{M_n} \varrho_{n,i} = 1. \quad (2)$$

Two types of weights $\varrho_{n,i}$ will be considered, that are deterministic and random, corresponding to the common and independent design, respectively. Let us defined the $(M_n \times M_n)$ invertible matrix

$$\mathcal{D}_n = \text{diag}(\varrho_{n,1}, \dots, \varrho_{n,M_n}).$$

¹For the prediction, the cases $M_n = 0$ and $M_n > 0$ can be investigated (keep in mind)

We introduce the operator $\Pi_{n,0} : \mathcal{H} \rightarrow \mathbb{R}^{M_n}$ which is defined as

$$\Pi_{n,0}(f) = \mathcal{D}_n^{1/2}(f(T_{n,1}), \dots, f(T_{n,M_n}))^\top, \quad f \in \mathcal{H}.$$

We also define $\Pi_{n,0}^* : \mathbb{R}^{M_n} \rightarrow \mathcal{H}$ and $\Pi_{n,1}^* : \mathbb{R}^{M_n} \rightarrow \mathcal{H}$ such that

$$[\Pi_{n,0}^* v](\cdot) = \sum_{i=1}^{M_n} v_i c_0(T_{n,i}, \cdot) \sqrt{\varrho_{n,i}} \quad \text{and} \quad [\Pi_{n,1}^* v](\cdot) = \sum_{i=1}^{M_n} v_i c_1(T_{n,i}, \cdot) \sqrt{\varrho_{n,i}}, \quad v = (v_1, \dots, v_{M_n})^\top \in \mathbb{R}^{M_n}.$$

Note that $[\Pi_{n,0}^* \Pi_{n,0} f](\cdot)$, $[\Pi_{n,1}^* \Pi_{n,0} f](\cdot)$, can be written under the matrix forms

$$[\Pi_{n,0}^* \Pi_{n,0} f](s) = C_{n,0}(t)^\top \mathcal{D}_n^{1/2} \Pi_{n,0} f, \quad [\Pi_{n,1}^* \Pi_{n,0} f](s) = C_{n,1}(s) \mathcal{D}_n^{1/2} \Pi_{n,0} f,$$

where

$$C_{n,0}(t) = (c_0(T_{n,1}, t), \dots, c_0(T_{n,M_n}, t))^\top = \mathcal{D}_n^{-1/2} \Pi_{n,0} c_0(\cdot, t),$$

and

$$C_{n,1}(t) = (c_0(T_{n,1}, t), \dots, c_1(T_{n,M_n}, t))^\top = \mathcal{D}_n^{-1/2} \Pi_{n,0} c_1(\cdot, t).$$

Furthermore, $\Pi_{n,0} \Pi_{n,0}^*$ is a linear operator from $\mathbb{R}^{M_n}$ to $\mathbb{R}^{M_n}$ which can be identified with the $(M_n \times M_n)$ -matrix

$$\mathcal{D}_n^{-1/2} \Pi_{n,0} \Pi_{n,0}^* \mathcal{D}_n^{-1/2} = (c_0(T_{n,i}, T_{n,j}))_{1 \leq i, j \leq M_n}.$$

With this new notation, rewrite the model (1) as,

$$\mathbf{Y}_n = \mathcal{D}_n^{-1/2} \Pi_{n,0} X_n + \sigma \epsilon_n \in \mathbb{R}^{M_n}, \quad n = 1 \dots N,$$

where $\mathbf{Y}_n = (Y_{n,1}, \dots, Y_{n,M_n})^\top$ is the stacked observation vector associated with the curve X_n , and $\epsilon_n = (\varepsilon_{n,1}, \dots, \varepsilon_{n,M_n})^\top$. Moreover, note that if Q_n converges weakly to the uniform distribution on $I = (0, 1]$ in probability, then for any continuous function f in $\mathcal{H}$ and any $s \in \mathcal{H}$,

$$\text{for } k \in \{0, 1\}, \quad [\Pi_{n,k}^* \Pi_{n,0} f] - (C_k f)(s) \rightarrow 0, \quad \text{in probability,}$$

where

$$[\Pi_{n,k}^* \Pi_{n,0} f](s) = \int_0^1 f(t) c_k(s, t) dQ_n(t) \quad \text{and} \quad (C_k f)(s) = \int_0^1 f(t) c_k(s, t) dt.$$

2.3 Regularized linear predictor

Let $n_0 \in \{1, \dots, N\}$ be fixed. We aim to constructing a predictor for the curve $\{X_{n_0+1}(u), u \in I\}$ using the observations of the L lagged curves, starting from n_0 . To simplify the presentation, we set $L = 1$. Let $\mathbf{Y}_{n_0} \in \mathbb{R}^{M_{n_0}}$ be the stacked observation vector associated with the curve X_{n_0} .

In the case of perfectly observed curves (in particular, $\sigma^2 = 0$), following [Bosq \(2000, Section 1.6\)](#), the Best Linear Unbiased Predictor of $X_{n_0+1} - \mu$ given $X_{n_0} - \mu$ is the orthogonal projection of $X_{n_0+1} - \mu$ onto the $\mathcal{L}$ -closed subspace,

$$\mathcal{G} = \{\psi(X_{n_0} - \mu), \quad \psi \in \mathcal{L}\}.$$

This predictor is the BLUP in the sense that it minimizes the mean squared error in $\mathbb{L}_{\mathcal{H}}^2$:

$$\psi \in \arg \min_{\phi \in \mathcal{L}} \mathbb{E} [\|X_{n_0+1} - \mu - \phi(X_{n_0} - \mu)\|_{\mathcal{H}}^2]. \quad (3)$$

The function ψ defining the projection and attaining the minimum in the optimization problem above can be shown to be a solution of the equation $C_1 = \psi C_0$ ([Hörmann and Kokoszka, 2012, Section 3.2](#)), where C_0 and C_1 are the lag-0 and lag-1 autocovariance operators, respectively. The operator C_0 is a Hilbert-Schmidt operator whose inverse is generally unbounded, leading to an ill-posed problem. Therefore, the solution in (3) cannot simply be defined as $\psi = C_0^{-1} C_1$.

Tikhonov regularization is a classical tool in the theory of ill-posed problems and has been applied to functional data analysis (Zhao, 2026, and references therein). The Tikhonov-regularized predictor of X_{n_0+1} in the case of perfectly observed curves is

$$\mathfrak{X}_{n_0+1}(t; \alpha) = \mu(t) + \{C_1(C_0 + \alpha \text{Id})^{-1}(X_{n_0} - \mu)\}(t), \quad t \in I,$$

where Id denotes the identity operator and $\alpha > 0$ is the regularization parameter. In Proposition 1, we introduce a sample-based approximation, $\mathfrak{X}_{n_0+1}^{(M_{n_0})}$, and study its uniform convergence in probability.

In the following, for a sample of design points $T_{n_0,i}$, $1 \leq i \leq M_{n_0}$, let $g_{n_0,i}(T_{n_0,i})$'s be the leave-one-out Parzen-Rosenblatt estimator computed at sample points, that is

$$\hat{g}_{n_0,i}(T_{n_0,i}) = \frac{1}{M_{n_0} - 1} \sum_{1 \leq j \leq M_{n_0}, j \neq i} K_h(T_{n_0,i} - T_{n_0,j}), \quad 1 \leq i \leq M_{n_0}.$$

Here, $K_h(\cdot) = K(\cdot/h)$ with $K(\cdot)$ an univariate kernel function and h the bandwidth. *****TBC assumption on this kernel and bandwidth *****

Proposition 1. *****TBC assumptions required here ***** $M_{n_0}h^{2r} = O_{\mathbb{P}}(1)$ ******* c_0 **uniform Hölder continuous ***** **perhaps take $r = 2$ to make things simple ***** Assume that, almost surely, the trajectories of X_n are β -Hölder continuous for some $\beta \in (0, 1]$, and

$$\mathbb{E}[\mathfrak{L}_n(\beta)^2] < \infty \quad \text{where} \quad \mathfrak{L}_n(\beta) = \sup_{u, v \in I, u \neq v} \frac{|X_n(u) - X_n(v)|}{|u - v|^\beta}.$$

Let Q_{n_0} be defined as in (2) and $\mathcal{D}_{n_0} = \text{diag}(\varrho_{n_0,1}, \dots, \varrho_{n_0,M_{n_0}})$ with

$$\varrho_{n_0,i} = \begin{cases} M_{n_0}^{-1} & \text{if equidistant common design} \\ \{M_{n_0}\hat{g}_{n_0,i}(T_{n_0,i})\}^{-1} & \text{if independent design.} \end{cases}$$

For $\alpha > 0$, consider the following Tikhonov regularized the predictor : $\forall t \in I$,

$$\mathfrak{X}_{n_0+1}^{(M_{n_0})}(t; \alpha, \sigma^2 \mathcal{D}_{n_0}) = \mu(t) + \left\{ \Pi_{n_0,1}^* (\Pi_{n_0,0} \Pi_{n_0,0}^* + \sigma^2 \mathcal{D}_{n_0} + \alpha I_{M_{n_0}})^{-1} (\mathcal{D}_{n_0}^{1/2} \mathbf{Y}_{n_0} - \Pi_{n_0,0} \mu) \right\}(t). \quad (4)$$

Then

$$\mathbb{E}_{M,T} \left[\sup_{t \in I} \left| \mathfrak{X}_{n_0+1}^{(M_{n_0})}(t; \alpha, \sigma^2 \mathcal{D}_{n_0}) - \mathfrak{X}_{n_0+1}(t; \alpha) \right| \right] = \alpha^{-2} O_{\mathbb{P}}(r(M_{n_0}; \beta)),$$

where

$$r(M_{n_0}; \beta) = \begin{cases} M_{n_0}^{-\beta} & \text{if } \varrho_{n_0,i} = M_{n_0}^{-1} \\ M_{n_0}^{-1/2} h^\beta + h^r + \log(1/h) M_{n_0}^{-1} h^{-1} & \text{if } \varrho_{n_0,i} = \{M_{n_0}\hat{g}_{n_0,i}(T_{n_0,i})\}^{-1}. \end{cases}$$

Remark 1. A marked distinction exists between the the approximation error rates of the theoretical Tikhonov-regularized predictor. With $r = 2$ (H3) and a second order kernel K , the best achievable rate in the independent design case (up to logarithm factors) is $M_{n_0}^{-1/2} h^{\beta/3} + M_{n_0}^{-2/3}$, which is faster than attainable under an equidistant common design if $\beta < 2/3$. Smoother curves X_n with $\beta > 2/3$ allow for smaller error rate in the common design case. It is important to note that in the independent design setting, even if the density g is given or can be estimated very accurately from other data sets, estimating it using the M_{n_0} design points corresponding to X_{n_0} is preferable, as this leads to an improved convergence rate.

Remark 2. It is worthwhile to notice that our predictor introduce in (4) becomes an usual BLUP (Best linear unbiased estimator) if $\alpha = 0$ (no regularization) and the $T_{n_0,i}$ are the points of an equidistant design. The usual BLUP value at a point $t \in I$ is solution of the problem

$$\min_{b \in \mathbb{R}^{M_{n_0}}} \mathbb{E}_{M,T} \left[\left\{ b^\top (\mathbf{Y}_{n_0} - \mathcal{D}_{n_0}^{-1/2} \Pi_{n_0,0} \mu) - (X_{n_0+1}(t) - \mu(t)) \right\}^2 \right].$$

See [Robinson \(1991\)](#). Our Proposition 1 implicitly implies that this type of predictor is inappropriate for prediction in FTS if the design points $T_{n_0,i}$ are random with a non-uniform density *and for equidistant common design*.

In practice, the unknown parameters involved in the construction of the predictor must be replaced by suitable estimators. Let $\hat{\mu}$, $\hat{c}_0$ and c_1 be uniformly consistent estimators of μ , c_0 and c_1 respectively. Then, the Tikhonov-regularized predictor is given, for every $t \in I$, by

$$\hat{\mathfrak{X}}_{n_0+1}^{(M_{n_0})}(t; \alpha, \sigma^2 \mathcal{D}_{n_0}) = \hat{\mu}(t) + \left\{ \hat{\Pi}_{n_0,1}^* \left(\Pi_{n_0,0} \hat{\Pi}_{n_0,0}^* + \hat{\sigma}^2 \mathcal{D}_{n_0} + \alpha I_{M_{n_0}} \right)^{-1} (\mathcal{D}_{n_0}^{1/2} \mathbf{Y}_{n_0} - \Pi_{n_0,0} \hat{\mu}) \right\}(t),$$

where

$$\hat{\Pi}_{n_0,\ell}^* : \mathbb{R}^{M_{n_0}} \mapsto \mathcal{H}, \quad v \mapsto \sum_{i=1}^{M_{n_0}} \sqrt{\rho_{n_0,i}} c_\ell(T_{n_0,i}, \cdot) v_i, \quad \ell = 0, 1.$$

Proposition 2. Under the assumptions of Theorems 1, 2, and 3, together with those of Proposition 1,

$$\begin{aligned} \|\hat{\mathfrak{X}}_{n_0+1}^{(M_{n_0})}(t; \alpha, \sigma^2 \mathcal{D}_{n_0}) - \mathfrak{X}_{n_0+1}^{(M_{n_0})}(t; \alpha, \sigma^2 \mathcal{D}_{n_0})\|_\infty &= \mathcal{O}_{\mathbb{P}} \left(\|\hat{\mu} - \mu\|_\infty (1 + \alpha^{-1} \|c_1\|_\infty) \right. \\ &\quad \left. + \sqrt{2 \|X_{n_0} - \mu\|_\infty^2 + 2\sigma^2} \left\{ \alpha^{-1} \|\hat{c}_1 - c_1\|_\infty + \alpha^{-2} \left(\|\hat{c}_0 - c_0\|_\infty + |\hat{\sigma} - \sigma| \max_i \rho_{n,i} \right) \|c_1\|_\infty \right\} \right) \end{aligned}$$

2.4 Choice of α

Let ψ an unknown Hilbert-Schmidt operator solution of $C_1 = \psi C_0$. Assume that,

(H6) The covariance operator C_0 is injective, compact and of trace class. Its eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots > 0$ admit the polynomial decay

$$c_1 k^{-a} \leq \lambda_k \leq c_2 k^{-a}$$

for some constants $0 < c_1 \leq c_2 < \infty$ and $a > 1$.

(H7) There exist constant $\gamma > 0$ and $\rho < \infty$ and a Hilbert-Schmidt operator F with $\|F\|_{HS} \leq \rho$ such that $\psi = F C_0^\gamma$ and $\|\psi\|_{op} < \infty$.

From [Zhao \(2026\)](#) (ils sont sous FAR(1)),

$$\|\mathfrak{X}_{n_0+1}(\cdot, \alpha) - (\mu + \psi(X_{n_0} - \mu))\|_{\mathcal{H}} \leq \|X_{n_0} - \mu\|_{\mathcal{H}} \mathcal{O}_{\mathbb{P}}(\alpha^{\min(\gamma,1)} + \alpha^{-1}/\sqrt{N})$$

where ψ is a solution of $C_1 = \psi C_0$ satisfying the source condition with constant γ

Equidistant common design

$$\alpha^{\min(\gamma,1)} + \alpha^{-1} (1/\sqrt{N} + \|\hat{\mu} - \mu\|_\infty + \|\hat{c}_1 - c_1\|_\infty) + \alpha^{-2} (M_{n_0}^{-\beta} + \|\hat{c}_0 - c_0\|_\infty + |\hat{\sigma} - \sigma| \max_i \rho_{n,i})$$

Independent design Choix de h

$$\alpha^{\min(\gamma,1)} + \alpha^{-1} (\sqrt{N} + \|\hat{\mu} - \mu\|_\infty + \|\hat{c}_1 - c_1\|_\infty) + \alpha^{-2} (M_{n_0}^{-1/2} h^\beta + \dots + \|\hat{c}_0 - c_0\|_\infty + |\hat{\sigma} - \sigma| \max_i \rho_{n,i})$$

2.5 Prediction interval

Next, we construct confidence bands with pointwise coverage for the trajectory X_{n_0} . We assume that both the stationary FTS and the measurement errors are Gaussian processes:

- (H8) The stationary FTS $\{X_n\}$ is a Gaussian process. That is, for any finite collection of indices $i_1, \dots, i_n \in \mathbb{Z}$, and any $t_1, \dots, t_n \in I$, the random vector $(X_{i_1}(t_1), \dots, X_{i_n}(t_n))^\top$ follows a multivariate normal distribution.
- (H9) The measurement errors $\{\varepsilon_{n,i}\}$ are Gaussian random variables.

In this setting, [Rubín and Panaretos \(2020\)](#) propose to build a prediction interval using the conditional distribution of $X_{n_0+1}(t)$ given $\mathbf{Y}_{n_0}$, which is a Gaussian distribution. But the expectation of this conditional distribution is the BLUP which diverges (C_0 is not injective), and its variance is ill-conditioned for large M_{n_0} . A such approach based on the conditional distribution is excluded.

Firts option: prediction interval based on the prediction error The aim is to characterize the distribution of $X_{n_0+1}(t) - \mathfrak{X}_{n_0+1}^{(M_{n_0})}(t; \alpha)$. Under the Gaussian assumption, this distribution is gaussian, its expectation is the bias of the predictor.

Second option: bootstrap interval [Paparoditis and Shang \(2023\)](#) propose bootstrap prediction interval based on the Karhunen–Loève representation. Maybe not appropriate in our settings.

Third option: conformal prediction This approach is distribution-free and does not require knowledge of the conditional distribution. It relies solely on the exchangeability of the observations $X_1, \dots, X_N$ to construct prediction intervals that are valid at a finite confidence level. The regularized predictor serves as a non-compliance score.

The conformal prediction is based on a simple principle : if $X_1, \dots, X_N$ are exchangeable, then for any non-conformity score $s(\cdot)$,

$$\frac{1}{N+1} \sum_{n=1}^N \mathbf{1}[s(X_n) \leq s(X_{N+1})]$$

is uniformly distributed on $\{1/N, \dots, 1\}$. This allows for the construction of valid prediction intervals at a finite level, without any assumptions about the distribution:

Exchangeability: $\{X_n\}$ is stationary, but not necessarily exchangeable.

Non-conformity score: a natural choice is

$$s_n(\alpha) = \sup_{t \in I} \left| X_n(t) - \mathfrak{X}_n^{(M_n-1)}(t; \alpha) \right|$$

that is, the sup-norm prediction error of the regularized predictor applied to each past observation.

Prediction interval: The empirical quantile of the scores gives the width of the prediction band.

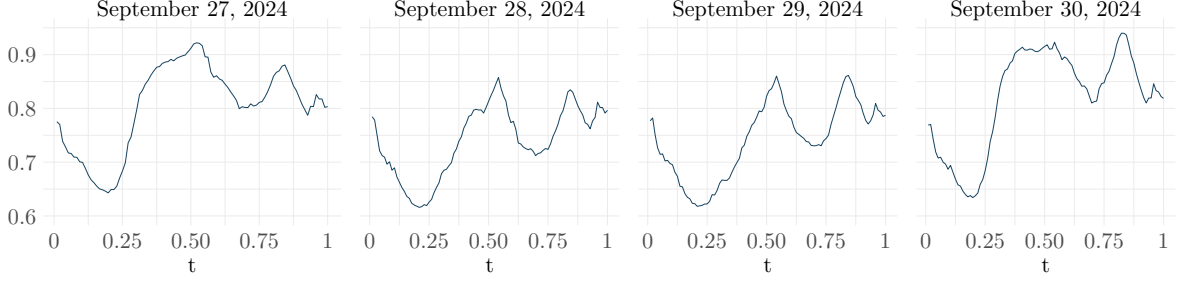


Figure 1: The last four daily national electricity consumption curves in France from the dataset, with values normalised to the interval $[0, 1]$.

3 Numerical study

In this section, we present a finite-sample Monte Carlo simulation study and a real data application. The simulation study is based on national electricity consumption curves for France, available from the ODRÉ platform ([Open Data Réseaux Énergies](https://odre.reseaux-energies.fr/), 2024). These curves are used to estimate the parameters of a first-order autoregressive process (FAR(1)), which are then used to generate synthetic curves for the Monte Carlo simulations. For the real data application, we analyse daily Facebook stock volume curves, available at <https://firststratedata.com/>. All results were obtained using the R package `adaptiveFTS` and supplementary code, both of which are publicly available at <https://github.com/hmaissoro>. An Epanechnikov kernel ($K(u) = (3/4)(1 - u^2)$ for $|u| \leq 1$, and 0 otherwise) is used in all experiments.

3.1 Simulation setting

The Monte Carlo simulation is based on France’s daily national electricity consumption curves from June to September 2024. The dataset contains 122 curves, one for each day within this period. Each curve represents electricity consumption recorded at 96 evenly spaced time points throughout the day, with measurements taken every 15 minutes. For readability, we normalise the curves to the interval $[0, 1]$ by dividing each value by the maximum consumption observed over the entire period. Figure 1 displays the last four consumption curves from the dataset.

To simulate FTS with patterns similar to those of the observed curves, we estimate the parameters of a FAR(1) model from the data and use them to generate synthetic FTS samples. We consider the FAR(1) process with an associated innovation process $\{\xi_n\}$, defined as:

$$X_n(u) = \mu(u) + \int_0^1 \psi(u, s)(X_{n-1}(s) - \mu(s)) ds + L_t \xi_n(u), \quad (5)$$

where the mean function μ and the kernel of the autoregressive operator ψ are estimated from the real consumption curves. The innovation process $\{\xi_n\}$ is independently generated as a multi-fractional Brownian motion, with the Hurst index function also estimated from the real consumption curves (see [Stoev and Taqqu, 2006](#)). Let us note that, as shown in Example 2 of [Maissoro et al. \(2025\)](#), if the kernel $\psi(\cdot, s)$ is Hölder continuous with a Hölder exponent greater than the regularity of $\{\xi_n\}$, then the FTS $\{X_n\}$ inherits the regularity of the innovation process. In particular, $\{X_n\}$ has a local Hölder exponent corresponding to the Hurst index function of $\{\xi_n\}$ and a Hölder constant L_t^2 .

The estimation procedure for μ and ψ using Fourier expansion is detailed in the supplementary material of [Maissoro et al. \(2025\)](#). The local Hölder exponent is estimated from the real dataset using the local regularity estimators proposed by [Maissoro et al. \(2025\)](#) over a grid of points and smoothed using natural cubic splines. Figure 2 shows the estimated local exponent $\hat{H}_t$, along with the estimated mean function μ and the autoregressive kernel ψ , where the kernel norm is normalised to 0.5.

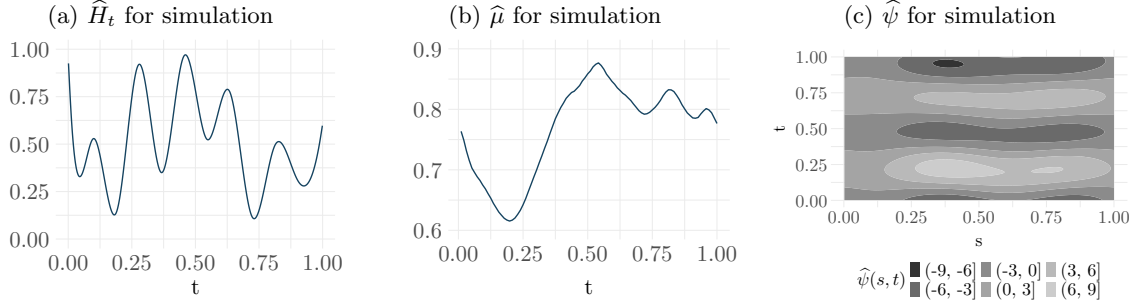


Figure 2: Simulation parameters estimated from consumption curves: (a) estimated local exponent function, (b) estimated mean function, (c) estimated autoregressive kernel.

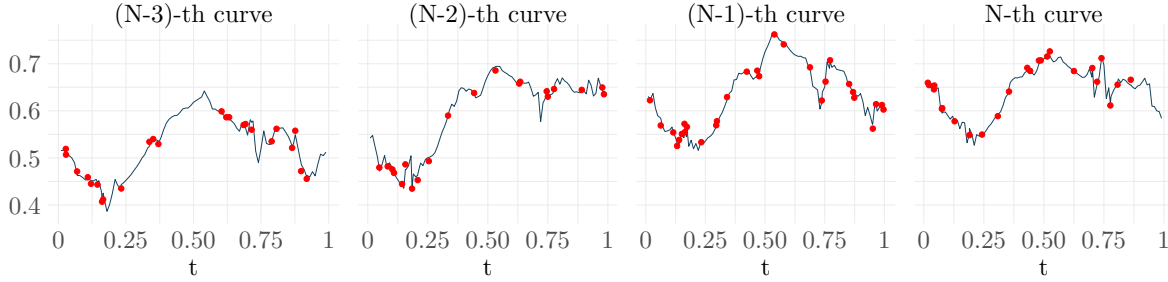


Figure 3: Last four simulated curves generated using the simulation setting with a sample size of $(N, \lambda) = (150, 25)$. The solid lines represent the sample paths, while the dots indicate the randomly observed noisy points on each curve, with an average of $\lambda = 25$ points per curve.

To obtain the data points according to (1), the number of points on each curve, M_n , is randomly generated from a uniform distribution between 0.8λ and 1.2λ , while the observation times $\{T_{n,i}\}$ are drawn independently from a uniform distribution on the interval $(0, 1]$. We set $L_t = 0.1$, corresponding to the mean of the estimates of the local Hölder constant of the consumption curves, evaluated over an equidistant grid of 99 points between 0.01 and 0.99, and obtained using the estimator proposed by Maissoro et al. (2025). For the observation error term, we set $\sigma(T_{n,i}) = 0.007$, corresponding to the mean standard deviation of the consumption curves estimated using (??) over the observation grid, and the $\varepsilon_{n,i}$ are centred Gaussian variables with unit variance. A sequence of 100 burn-in steps is used to initialise (5). Figure 3 illustrates the simulated sample paths and the randomly observed points for a sample size of $(N, \lambda) = (150, 25)$.

Finally, for the (auto)covariance study we generate different samples of FTS, whereas for the BLUP study we generate a single sample of FTS and regenerate its last curve. The aim of the (auto)covariance study is to show that the proposed method performs at least as well as the approach of Maissoro et al. (2025), which uses a single bandwidth. Meanwhile, the aim of the BLUP study is to show that a curve, here the last curve, is accurately reconstructed.

3.2 Implementation details

The estimation of the lag- ℓ , $\ell \geq 0$, (auto)covariance using the estimators in (??), (??), and (??) involves selecting optimal bandwidths based on the data-driven rules given in (??), (??), and (??), respectively. Computing $\hat{R}_{\Gamma_\ell}$ requires estimating the dependence coefficients $\mathbb{D}_\ell(s, t; h_s, h_t)$, $\mathbb{D}_\ell(t, h_t | s, h_s)$, and $\mathbb{D}(t; h_t)$, which is computationally intensive. To reduce computation time, we adopt a simplified approach. Let $\tilde{X}_n$ denote the NW estimator of the curve X_n , obtained using a bandwidth selected via cross-validation. Given a set of sample points $\{(Y_{n,1}, T_{n,1}), \dots, (Y_{n,M_n}, T_{n,M_n})\}$ for a curve X_n , the cross-validated optimal

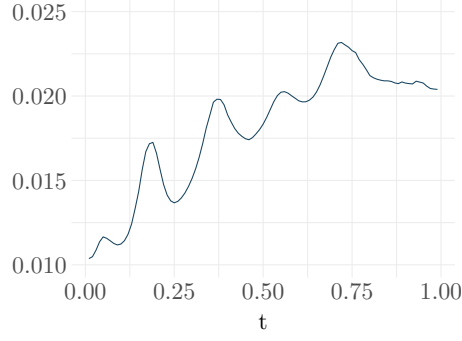


Figure 4: Variance of the constructed FTS process from (5) and the consumption curves.

bandwidth for the estimator in (??) is defined as

$$h^* \in \arg \min_h \sum_{i=1}^{M_n} \left\{ Y_{n,i} - \tilde{X}_n^{(-i)}(T_{n,i}; h) \right\}^2, \quad (6)$$

where $\tilde{X}_n^{(-i)}(T_{n,i}; h)$ denotes the estimator in (??) computed without the i -th observation and using bandwidth h . Moreover, let $\tilde{Z}_n(t) = \tilde{X}_n(t) - \tilde{\mu}_N(t)$, and define, for all $s, t \in I$,

$$\tilde{\mu}_N(t) = \frac{1}{N} \sum_{n=1}^N \tilde{X}_n(t), \quad \tilde{\Gamma}_\ell(s, t) = \frac{1}{N-\ell} \sum_{n=1}^{N-\ell} \tilde{Z}_n(s) \tilde{Z}_{n+\ell}(t), \quad \ell \geq 1.$$

Then, instead of estimating $\mathbb{D}_\ell(s, t; h_s, h_t)$, $\mathbb{D}_\ell(t, h_t | s, h_s)$, and $\mathbb{D}(t; h_t)$, we simply consider the following

$$\begin{aligned} \hat{\mathbb{D}}_\ell(s, t; h_s, h_t) &= \frac{1}{N-\ell} \sum_{n=1}^{N-\ell} \left\{ \tilde{Z}_n(s) \tilde{Z}_{n+\ell}(t) - \tilde{\Gamma}_\ell(s, t) \right\}^2 \\ &\quad + \sum_{k=1}^{N-\ell-1} \frac{2P_{N,\ell}^{-1}(s, t; h_s, h_t)}{N-\ell-k} \left| \sum_{n=1}^{N-\ell-k-1} \left\{ \tilde{Z}_n(s) \tilde{Z}_{n+\ell}(t) - \tilde{\Gamma}_\ell(s, t) \right\} \left\{ \tilde{Z}_{n+k}(s) \tilde{Z}_{n+\ell+k}(t) - \tilde{\Gamma}_\ell(s, t) \right\} \right|, \\ \hat{\mathbb{D}}_\ell(t, h_t | s, h_s) &= 24 \sum_{1 \leq n_1 < n_2 < n_3 \leq \ell_{\max}} (N-n_3)^{-1} \left| \sum_{k=1}^{N-n_3} \frac{p_{k,n_1,n_2}(t; h_t)}{P_{N,\ell}^3(s, t; h_s, h_t)} \left\{ \tilde{Z}_k \tilde{Z}_{k+n_1} \tilde{Z}_{k+n_2} \tilde{Z}_{k+n_3} \right\}(t) \right| \\ &\quad + 6 \sum_{1 \leq n_1 < n_2 \leq \ell_{\max}} (N-n_2)^{-1} \left| \sum_{k=1}^{N-n_2} \frac{p_{k,n_1,n_2,n_3}(t; h_t)}{P_{N,\ell}^3(s, t; h_s, h_t)} \left\{ \tilde{Z}_k^2 \tilde{Z}_{k+n_1} \tilde{Z}_{k+n_2} + \tilde{Z}_k \tilde{Z}_{k+n_1}^2 \tilde{Z}_{k+n_2} + \tilde{Z}_k \tilde{Z}_{k+n_1} \tilde{Z}_{k+n_2}^2 \right\}(t) \right|, \\ \hat{\mathbb{D}}(t; h_t) &= 24 \sum_{1 \leq n_1 < n_2 < n_3 \leq \ell_{\max}} (N-n_3)^{-1} \left| \sum_{k=1}^{N-n_3} \frac{p_{k,n_1,n_2}(t; h_t)}{P_N^3(t; h_t)} \left\{ \tilde{Z}_k \tilde{Z}_{k+n_1} \tilde{Z}_{k+n_2} \tilde{Z}_{k+n_3} \right\}(t) \right| \\ &\quad + 6 \sum_{1 \leq n_1 < n_2 \leq \ell_{\max}} (N-n_2)^{-1} \left| \sum_{k=1}^{N-n_2} \frac{p_{k,n_1,n_2,n_3}(t; h_t)}{P_N^3(t; h_t)} \left\{ \tilde{Z}_k^2 \tilde{Z}_{k+n_1} \tilde{Z}_{k+n_2} + \tilde{Z}_k \tilde{Z}_{k+n_1}^2 \tilde{Z}_{k+n_2} + \tilde{Z}_k \tilde{Z}_{k+n_1} \tilde{Z}_{k+n_2}^2 \right\}(t) \right|, \end{aligned}$$

where $N \gg \ell_{\max} = 3$ to avoid $\mathcal{O}(N^4)$ or $\mathcal{O}(N^3)$ complexity, and

$$\begin{aligned} p_{k,n_1,n_2}(t; h_t) &= \pi_k(t; h_t) \pi_{k+n_1}(t; h_t) \pi_{k+n_2}(t; h_t), \\ p_{k,n_1,n_2,n_3}(t; h_t) &= \pi_k(t; h_t) \pi_{k+n_1}(t; h_t) \pi_{k+n_2}(t; h_t) \pi_{k+n_3}(t; h_t). \end{aligned}$$

3.3 Autocovariance function estimation

We consider the pairs $(N, \lambda) \in \{150, 300, 600\} \times \{25, 50, 100, 200\}$ and generate $R = 400$ independent FTS. For each sample, the aim is to estimate the autocovariance using both the single-bandwidth estimator proposed by Maissoro et al. (2025) and our new estimator, which uses two bandwidth parameters. Hereafter, we refer to the estimator of Maissoro et al. (2025) as MPV24. The adaptive estimation of the autocovariance function via the estimator given in (??) involves selecting optimal bandwidths by minimising the risk function defined in (??). First, we consider the estimation of this risk function and then present a performance comparison with the MPV24 estimator.

To this end, we consider four pairs (s, t) and compute the average of the lag-1 autocovariance risk function $\widehat{R}_{\Gamma_1}(s, t; h_s, h_t)$, as defined in (??), over a bandwidth grid $\mathcal{H}_N \times \mathcal{H}_N$ based on the $R = 400$ FTS replications. Figure 5 shows the mean of the 400 replications of the estimates of the lag-1 autocovariance risk function for $(N, \lambda) = (150, 100)$ at four locations (s, t) . The estimates of the risk function are a convex function of (h_s, h_t) for all pairs (s, t) . Furthermore, since the regularity varies along the domain (see Figure 2-(a)), as expected, the minima bandwidth pairs $(h_1^*(s|t), h_1^*(t|s))$ are reached such that $h_1^*(s|t) \neq h_1^*(t|s)$, when $H_s \neq H_t$. Moreover, the difference between the selected bandwidths is greater for pairs (s, t) such that $|H_s - H_t|$ is large.

We conclude this section with a performance comparison of the new autocovariance estimator against the MPV24 estimator, which uses a single bandwidth. Let $\widehat{\Gamma}_{N,\ell}^{(i)}$ denote our new estimator computed from the i -th replication ($i = 1, \dots, R = 400$), and let $\widehat{\Gamma}_{\text{MPV24},\ell}^{(i)}$ denote the corresponding MPV24 estimator. For benchmarking, we use the empirical infeasible estimator, denoted by $\widehat{\Gamma}_{N',\ell}$, which is computed on a large sample of size N' . This estimator is feasible in simulation settings. That is,

$$\widehat{\Gamma}_{N',\ell}(s, t) = \frac{1}{N' - \ell} \sum_{n=1}^{N' - \ell} \{X_n(s) - \widehat{\mu}_{N'}(s)\} \{X_{n+\ell}(t) - \widehat{\mu}_{N'}(t)\}, \quad \text{where} \quad \widehat{\mu}_{N'}(t) = \frac{1}{N'} \sum_{n=1}^{N'} X_n(t).$$

Then, for each $i = 1, \dots, R$, we compute the pointwise error ratio:

$$\forall (s, t) \in I^2, \quad E_\ell^{(i)}(s, t) = \frac{\left| \widehat{\Gamma}_{N,\ell}^{(i)}(s, t; h_\ell^*(s|t), h_\ell^*(t|s)) - \widehat{\Gamma}_{N',\ell}(s, t) \right|}{\left| \widehat{\Gamma}_{\text{MPV24},\ell}^{(i)}(s, t; h^*) - \widehat{\Gamma}_{N',\ell}(s, t) \right|}. \quad (7)$$

With $N' = 5000$, we compute the error ratios $\{E_1^{(i)}(s, t), i = 1, \dots, R\}$ for the lag-1 autocovariance estimator at four locations (s, t) . Figure 6 presents the corresponding boxplots across all (N, λ) configurations. As anticipated from Theorem ??, the new estimator significantly outperforms MPV24 when the local regularities H_s and H_t differ, with greater gains observed for pairs (s, t) such that $|H_s - H_t|$ is large. For instance, the largest improvements are found at $(s, t) = (0.2, 0.3)$ and $(0.5, 0.2)$, where $(H_{0.2}, H_{0.3}) = (0.199, 0.844)$ and $(H_{0.5}, H_{0.2}) = (0.784, 0.199)$, respectively. In contrast, at $(s, t) = (0.8, 0.5)$, where $(H_{0.8}, H_{0.5}) = (0.445, 0.784)$, the two estimators tend to perform similarly as the sample size increases.

3.4 Covariance function estimation

Our new adaptive covariance estimator involves selecting optimal bandwidths by minimising both the estimates of the bivariate risk function (??) outside the diagonal band and the estimates of the univariate risk function given in (??) for the diagonal line estimator. We consider the same scenario and generated data as in the autocovariance estimation study to illustrate the performance of our new covariance estimator.

First, regarding the risk function estimation, Figure 7 shows the average over the 400 replications of the estimates of the covariance risk function (??) for $(N, \lambda) = (150, 100)$ at four locations. These plots provide evidence that the risk function is convex and can thus be effectively minimised to obtain optimal

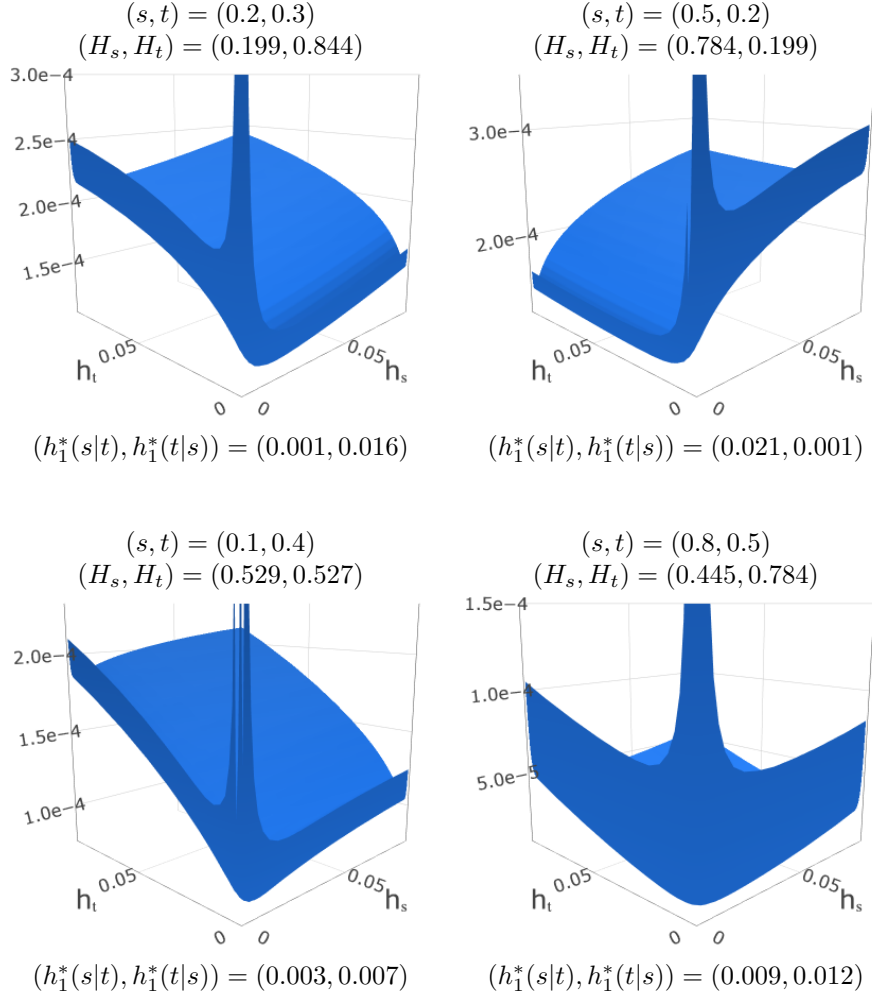


Figure 5: Empirical average of the risk function $\hat{R}_{R_1}(s, t; h_s, h_t)$ at $(s, t) \in \{(0.2, 0.3), (0.5, 0.2), (0.1, 0.4), (0.8, 0.5)\}$ computed over 400 independent replications of the FTS for the sample size $(N, \lambda) = (150, 100)$. For each (s, t) , the optimal bandwidths $(h_1^*(s|t), h_1^*(t|s))$ corresponding to the plotted risk function are indicated at the bottom of the respective panel.

bandwidth pairs. Similarly, Figure 8 illustrates that the risk function for the diagonal line is also convex, allowing straightforward selection of the optimal bandwidth for its estimation.

Furthermore, for the bivariate risk function (??), since the regularity varies across the domain (see Figure 2-(a)), the optimal bandwidth pairs $(h_0^*(s|t), h_0^*(t|s))$ are typically such that $h_0^*(s|t) \neq h_0^*(t|s)$ when $H_s \neq H_t$. The difference between the selected bandwidths becomes more pronounced for pairs (s, t) with large differences in local regularity, that is, when $|H_s - H_t|$ is large.

We conclude this section with a performance comparison between our proposed covariance estimator and its version that selects the same bandwidth for both arguments of the function. The error in (7) is then computed with respect to that covariance estimator instead of to the MPV24 estimator. For simplicity, we do not restate the formula. Figure 9 shows boxplots of the error ratio of the covariance estimator for four (s, t) pairs across different sample size configurations (N, λ) . As anticipated from Theorem ??, our estimator significantly outperforms the covariance estimator that uses the same local optimal bandwidth for both arguments, when the local regularities H_s and H_t differ, and the gain increases with larger values of $|H_s - H_t|$.

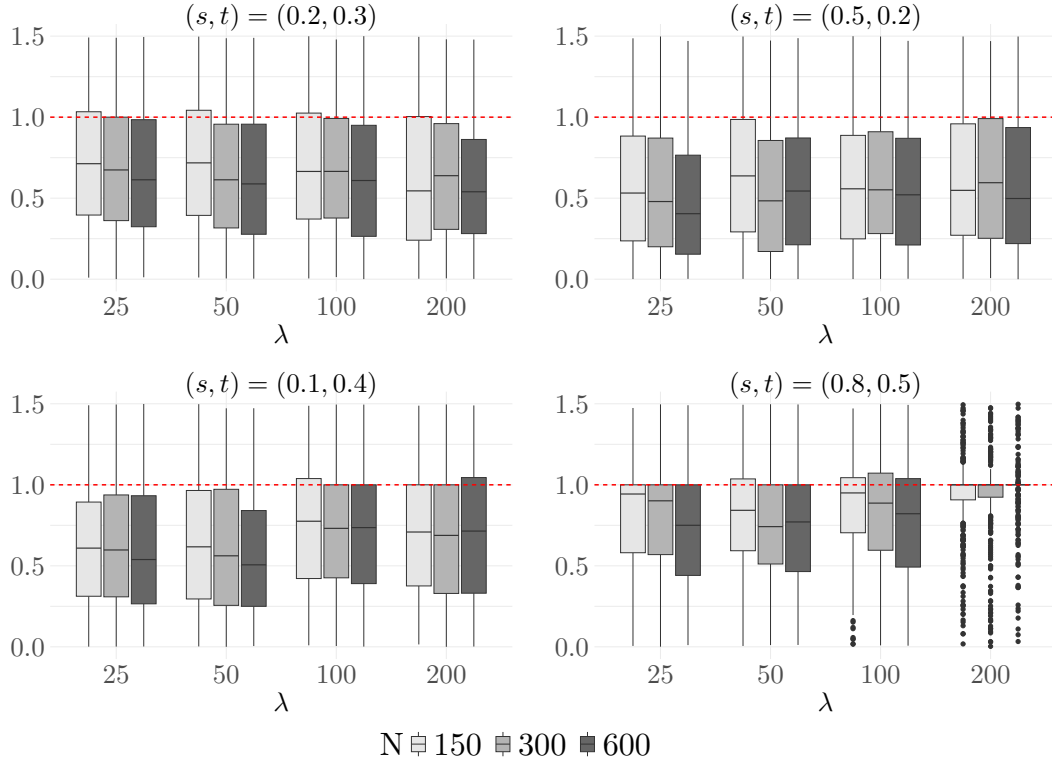


Figure 6: Boxplots of the error ratio (7) of the lag-1 autocovariance estimator for four pairs $(s, t) \in \{(0.2, 0.3), (0.5, 0.2), (0.1, 0.4), (0.8, 0.5)\}$. For each pair, the corresponding values of the local exponent function are $(H_{0.2}, H_{0.3}) = (0.199, 0.844)$, $(H_{0.5}, H_{0.2}) = (0.784, 0.199)$, $(H_{0.1}, H_{0.4}) = (0.529, 0.527)$, and $(H_{0.8}, H_{0.5}) = (0.445, 0.784)$.

3.5 Adaptive curve prediction

In this section, we study the performance of the adaptive BLUP through Monte Carlo simulations. Since the adaptive BLUP leverages information from both the current curve and the lagged curve, we first examine the empirical autoregressive effect and then present the Monte Carlo results.

3.5.1 Empirical autoregressive effect

Recall that the adaptive BLUP depends on adaptive estimates of the mean, covariance, and lag-1 autocovariance functions. To describe the strength of dependence between the current and lagged curves, we compute a functional analogue of the autocorrelation function for real-valued time series. The Functional Autocorrelation Function (FACF) at lag ℓ , considered by Horváth et al. (2016), is defined as

$$\hat{\rho}_\ell = \|\hat{\Gamma}_{N,\ell}\| \left[\int \hat{\Gamma}_{N,0}(t, t) dt \right]^{-1}, \quad (8)$$

where $\hat{\Gamma}_{N,\ell}$ estimates Γ_ℓ , and $\|\cdot\|$ denotes the standard norm in $\mathbb{L}_{\mathcal{H}}^2$. Moreover, $0 \leq \hat{\rho}_\ell \leq 1$. The FACF also serves as an empirical check for the presence of an autoregressive effect in the FTS. In fact, if there were no autoregressive effect, the FACF would equal zero, implying that the autocovariance-related top-right and bottom-left blocks in the matrix $\mathbb{V}_{n_0}$ would be nil and thus contribute nothing to the adaptive BLUP.

We use the same generated sample of FTS as in the study of the autocovariance function, estimate the lag-1 FACF for each of the 400 FTS replications using (8), and display the resulting histograms in

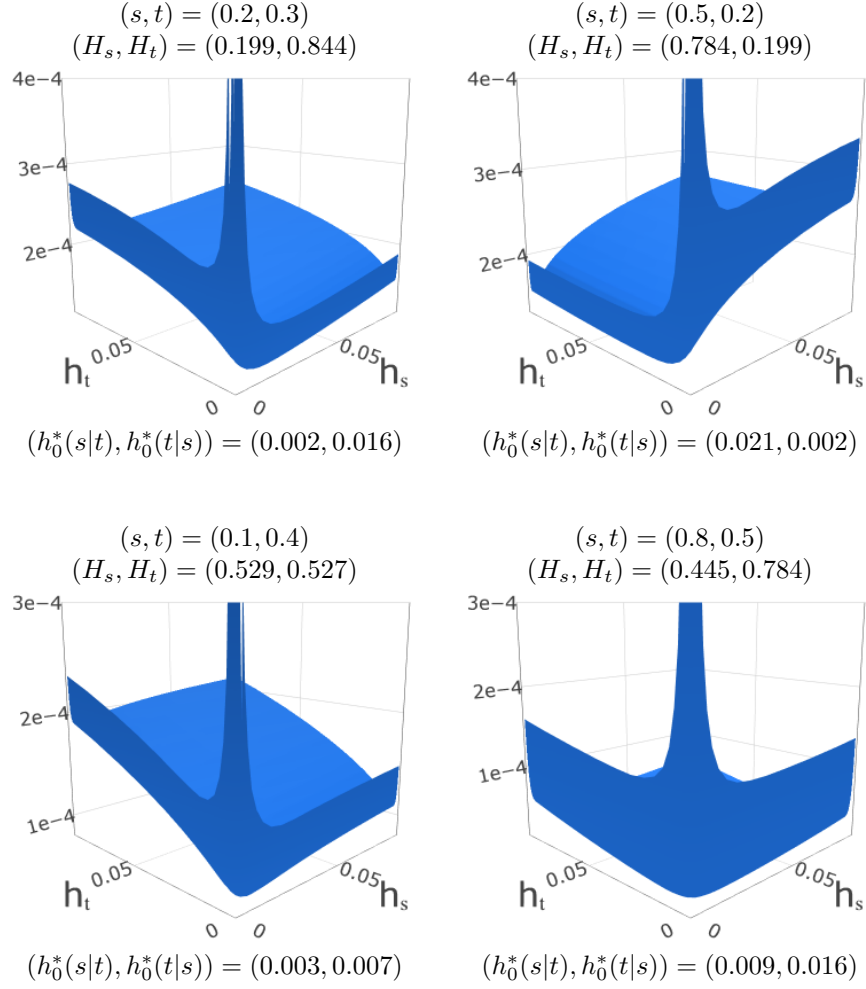


Figure 7: Empirical average of the risk function $\hat{R}_{\Gamma_0}(s, t; h_s, h_t)$ at $(s, t) \in \{(0.2, 0.3), (0.5, 0.2), (0.1, 0.4), (0.8, 0.5)\}$ computed over 400 independent replications of the FTS for the sample size $(N, \lambda) = (150, 100)$. For each (s, t) , the optimal bandwidths $(h_0^*(s|t), h_0^*(t|s))$ corresponding to the plotted risk function are indicated at the bottom of the respective panel.

Figure 10 for different sample sizes (N, λ) . The plots provide empirical evidence that the autocovariance-related blocks in $\mathbb{V}_{n_0}$ are not nil. Moreover, for our simulated FAR(1) model, most of the lag-1 FACF values are concentrated between 0.2 and 0.4.

3.5.2 Monte Carlo study

We now turn to the study of the predictive performance of the adaptive BLUP through Monte Carlo simulations. Given a sample of FTS, the objective is to assess the prediction performance of its final curve, that is, $n_0 = N$, under various scenarios depending on the available observation points for that curve. To this end, we adopt a simulation setup that differs slightly from the one used in the study of the (auto)covariance function. We consider the pairs $(N, \lambda) \in \{150, 300, 600\} \times \{25, 50, 100\}$, generate a single sample of FTS, and regenerate its final curve $R = 400$ times. The prediction accuracy of each regenerated final curve is then evaluated, given all or only a subset of its observed points, while maintaining the other curves of the FTS sample unchanged.

Table 1 shows the bias and standard deviation of the estimates of $\hat{\mathbf{x}}_{n_0}(t)$ obtained for the prediction

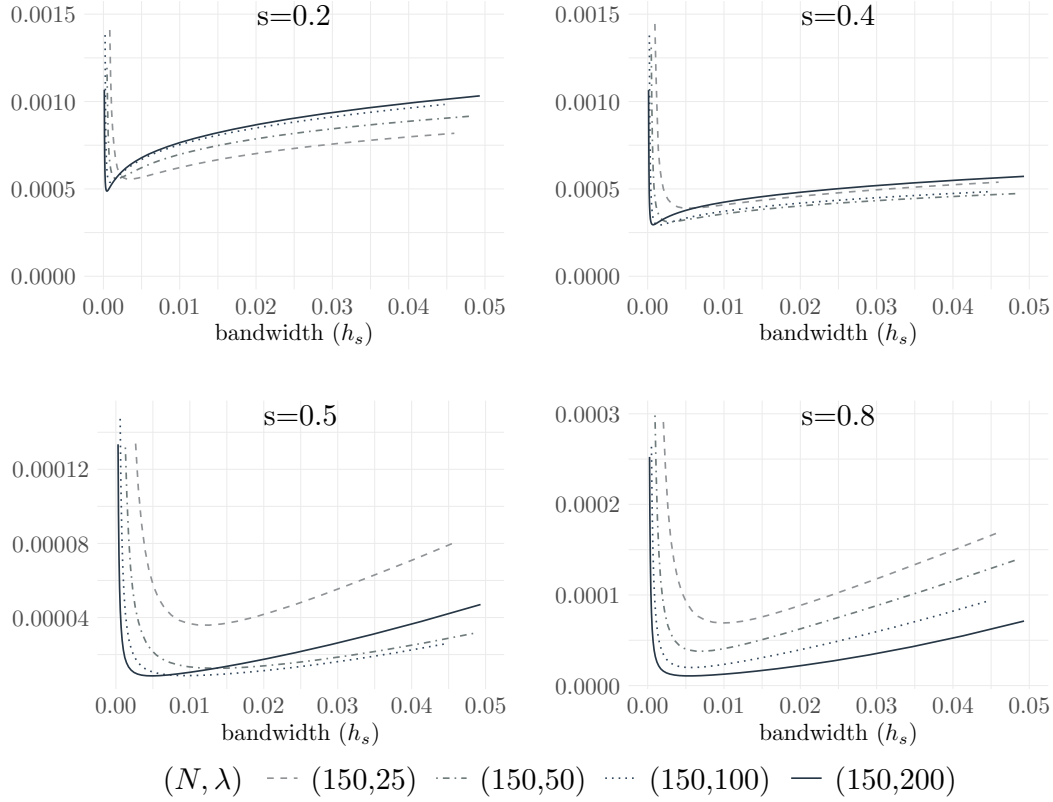


Figure 8: Empirical average of the risk function $\widehat{R}_\mu(s, h_s)$ at $s \in \{0.2, 0.4, 0.5, 0.8\}$, computed over 400 independent replications of the FTS for sample sizes $(N, \lambda) \in \{(150, 25), (150, 50), (150, 100), (150, 200)\}$.

N	λ	$t = 0.2$		$t = 0.4$		$t = 0.7$		$t = 0.8$	
		Bias	Sd	Bias	Sd	Bias	Sd	Bias	Sd
150	25	0.0229	0.1173	-0.0187	0.1209	0.0331	0.1404	0.0001	0.1275
150	50	0.0430	0.1033	0.1017	0.1293	-0.0050	0.1194	-0.0093	0.1285
150	100	-0.0085	0.1006	0.0672	0.0994	0.0232	0.1326	-0.0207	0.1005

Table 1: Bias and standard deviation (SD) of adaptive BLUP estimates for the last curve, based on 400 independent replications.

of the last curve, when all observation points $\{(Y_{n_0,i}, T_{n_0,i}), 1 \leq i \leq M_{n_0}\}$ are available. As expected, both the bias and the variance of the adaptive BLUP decrease as the sample size increases. Notably, the estimated standard deviation tends to increase with t . This is partly due to the sample paths becoming rougher towards the right end of the domain I , as illustrated by the shape of the local exponent function in Figure 2-(a). Moreover, larger values of t are associated with higher marginal variance $\text{Var}(X_t)$, as shown in Figure 4. These factors lead to less accurate estimates of the mean and (auto)covariance functions, and consequently to less accurate predictions by the adaptive BLUP.

We now compare our prediction procedure to that of [Rubín and Panaretos \(2020\)](#), referred to as RP20, as well as to the Nadaraya–Watson (NW) estimator using a cross-validated bandwidth as defined in (6). The comparison with the NW estimator is limited to the scenario where all observation points on the last curve are available. For the comparison with RP20, we also consider three additional scenarios in

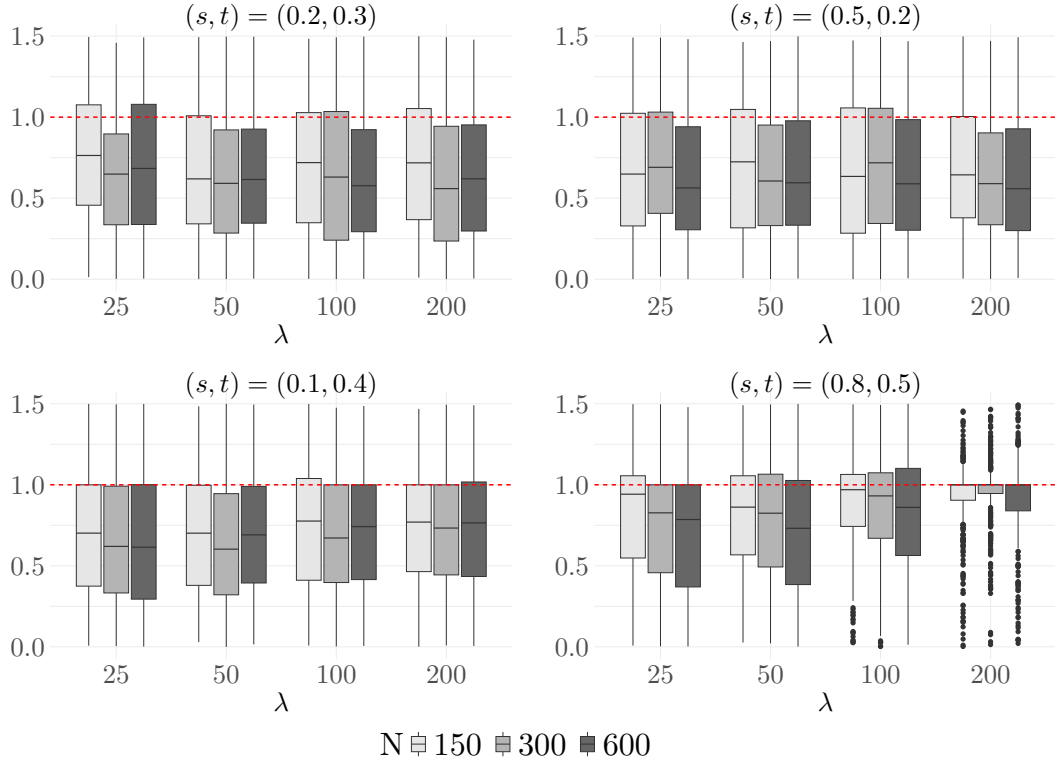


Figure 9: Boxplot of the error ratio (7) of the covariance estimator for four pairs $(s, t) \in \{(0.2, 0.3), (0.5, 0.2), (0.1, 0.4), (0.8, 0.5)\}$. For each pair, the corresponding values of the local exponent function are $(H_{0.2}, H_{0.3}) = (0.199, 0.844)$, $(H_{0.5}, H_{0.2}) = (0.784, 0.199)$, $(H_{0.1}, H_{0.4}) = (0.529, 0.527)$, and $(H_{0.8}, H_{0.5}) = (0.445, 0.784)$.

which the available observation points for the last curve lie within the subdomains $(0, 1/4]$, $(0, 1/2]$, and $(0, 3/4]$.

Let us denote by

$$\mathcal{T}_{n_0}^\alpha = \{(Y_{n,i}, T_{n,i}) : T_{n,i} \in (0, \alpha], 1 \leq i \leq M_n, 1 \leq n \leq N\}, \quad \alpha = 1/4, 1/2, 3/4,$$

the set of all available observation points in the FTS, with the last curve observed only over $(0, \alpha]$. Predictions based on the data $\mathcal{T}_{n_0}^\alpha$ are compared using the ratio of integrated squared errors (ISE). Let the superscript (k) denote the k -th replication of the last curve. Then, the ISE ratio of the adaptive BLUP relative to a given method **Meth** (either NW or RP20) is defined as

$$\text{Ratio}^{(k)}(\text{Meth}) = \frac{\int_0^1 \left(\widehat{\mathbf{x}}_{n_0}^{(k)}(t) - X_{n_0}^{(k)}(t) \right)^2 dt}{\int_0^1 \left(\widehat{\mathbf{x}}_{\text{Meth}}^{(k)}(t) - X_{n_0}^{(k)}(t) \right)^2 dt}.$$

Recall that the RP20 method is a BLUP-based approach, where the mean is estimated using a local linear smoother and the lag- ℓ (auto)covariance function is estimated using a local linear surface smoother (off-diagonal, when $\ell = 0$) and a local quadratic smoother (on the diagonal, when $\ell = 0$). Smoothing bandwidths are selected via K -fold cross-validation coupled with Bayesian optimisation (implemented in MATLAB). Observations are randomly sampled on a regular grid of 241 points, and B-spline projections are used to reduce the computational time during optimisation.

Figure 11 presents boxplots of the logarithm of the ISE ratios of the adaptive BLUP relative to the NW estimator. These plots show that the adaptive BLUP generally outperforms the NW estimator,

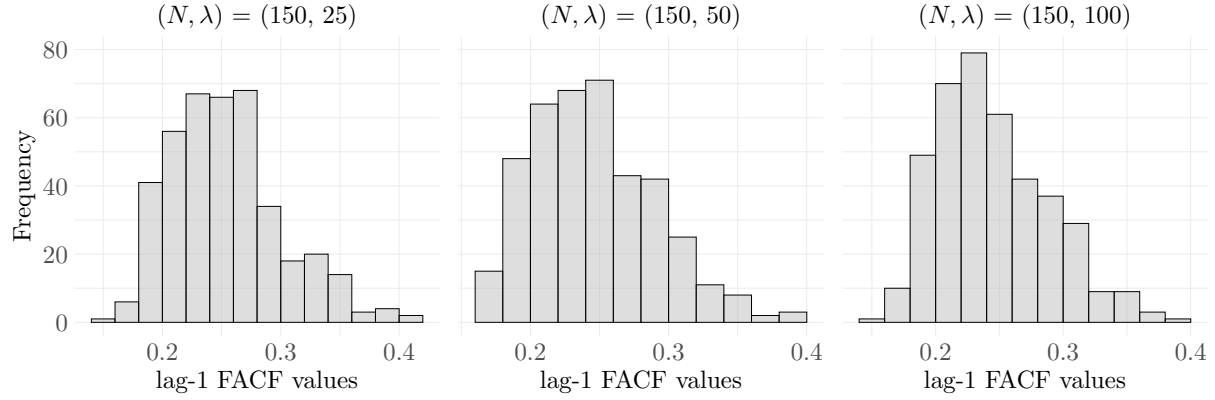


Figure 10: Histograms of estimated lag-1 FADF values for different (N, λ) configurations.

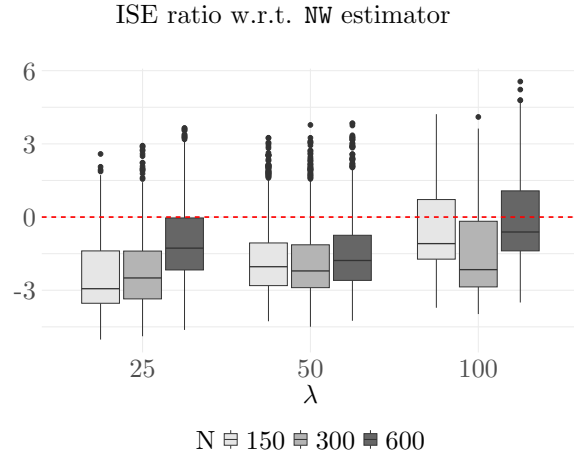


Figure 11: Boxplots of the log ISE ratios of the adaptive BLUP relative to the NW estimator, in the scenario where all observation points of the last curve to be predicted are available.

with the log-relative gain tending to zero as λ increases. When $\lambda \in \{25, 50\}$, the adaptive BLUP clearly provides better performance. However, for $\lambda = 100$, the NW estimator becomes more accurate and tends to perform similarly to the adaptive BLUP. This is likely due to the decrease in the mean squared error of the NW estimator as λ increases, which improves its ability to reconstruct the curves effectively.

Figure 12 shows the boxplots of the log ISE ratios of the adaptive BLUP relative to the RP20 method under the four scenarios $\alpha = 1, 1/4, 1/2$, and $3/4$. In the full-observation case ($\alpha = 1$), the adaptive BLUP outperforms RP20 when $\lambda = 25$, although this advantage diminishes as N increases. However, for $\lambda \in \{50, 100\}$, RP20 yields better performance than the adaptive BLUP (see Figure 12-(a)). In contrast, in the partial-observation scenarios with $\alpha \in \{1/4, 1/2, 3/4\}$, the adaptive BLUP consistently outperforms RP20 across all considered combinations of (N, λ) , as illustrated in Figures 12-(b), (c), and (d), respectively.

3.6 Real data application

We consider the prediction of daily Facebook stock volume curves. The dataset, available at <https://firststratedata.com/>, contains one-minute time-frame records of the stock price and volume. We extract observations from 1 January 2021 to 31 March 2022, using only the market's regular trading

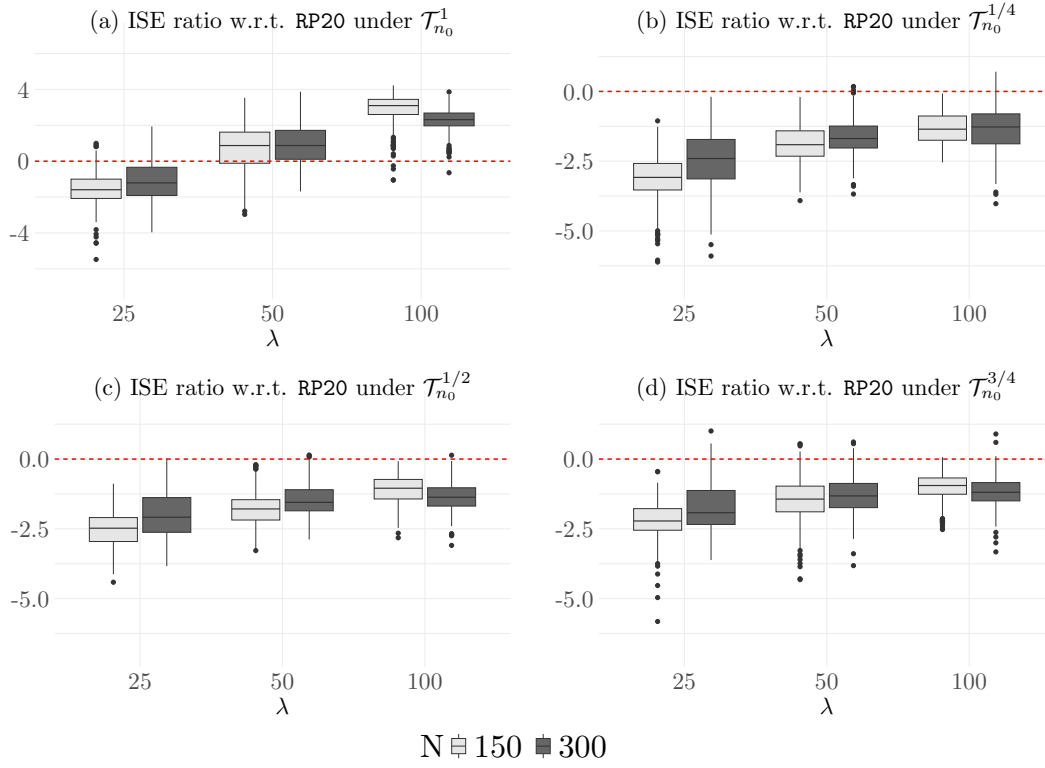


Figure 12: Boxplots of the log ISE ratios of the adaptive BLUP relative to RP20 for predicting the last curve, with available observations in $(0, 1]$, $(0, 1/4]$, $(0, 1/2]$, and $(0, 3/4]$, respectively.

hours, from 9:30 a.m to 4:00 p.m.

We construct daily stock return curves as the difference between the logarithm of the price at each time during the day and the opening price at 9:30 a.m. We are interested in predicting the traded volume when the absolute value of the return exceeds 0.5%. Thus, for each day, we retain only the stock volume observations where the absolute return is greater than 0.5%. The resulting daily volume curves are observed at random times between 9:30 a.m. and 4:00 p.m., with a maximum of 390 points per curve, corresponding to the total number of minutes during regular trading hours. The observation times are normalised to the interval $(0, 1]$. It is worth noting that traded volume is typically very high near the market's close; to mitigate the impact of extreme values, we exclude any volume observations above the 97.5th percentile of the entire sample.

Figure 13 shows the last four curves in the time series. Note that, due to the return threshold, a curve may have no observations satisfying the condition on a given day. Such cases account for fewer than 10% of all days and, therefore, we neglect the effect of this missingness. The final dataset consists of 284 curves with an average of 282 points per curve.

To assess the empirical autoregressive structure, we estimate the lag-1 FACF using (8), obtaining a value of 0.31. This provides evidence that the autocovariance-related blocks in $\mathbb{V}_{n_0}$ are non-negligible, indicating that the lagged curve contributes meaningfully to the adaptive BLUP. Moreover, this estimated FACF falls within the empirical distribution of lag-1 FACF values observed in our simulation study (see Figure 10).

Figure 14 presents a comparison between the adaptive BLUP predictions and the true stock volume curve to be predicted, together with a 95% pointwise confidence interval. The comparison is performed under three scenarios, in which the available observations for the curve to be predicted are restricted to the interval $(0, \alpha]$ with $\alpha \in \{1/4, 1/2, 3/4\}$. Across these scenarios, the adaptive BLUP effectively

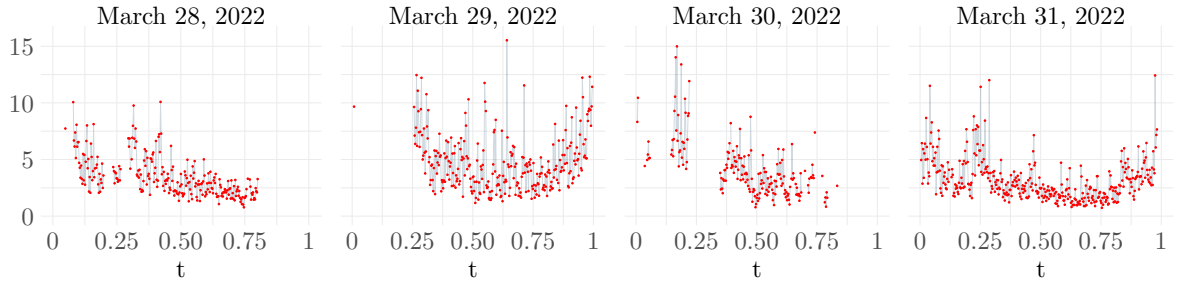


Figure 13: Observation points for the last four daily Facebook stock volume curves in the dataset, with values normalised by 10^4 . Dots indicate the observation points, while the shaded lines illustrate the pattern between them.

captures local irregularities and demonstrates satisfactory predictive performance.

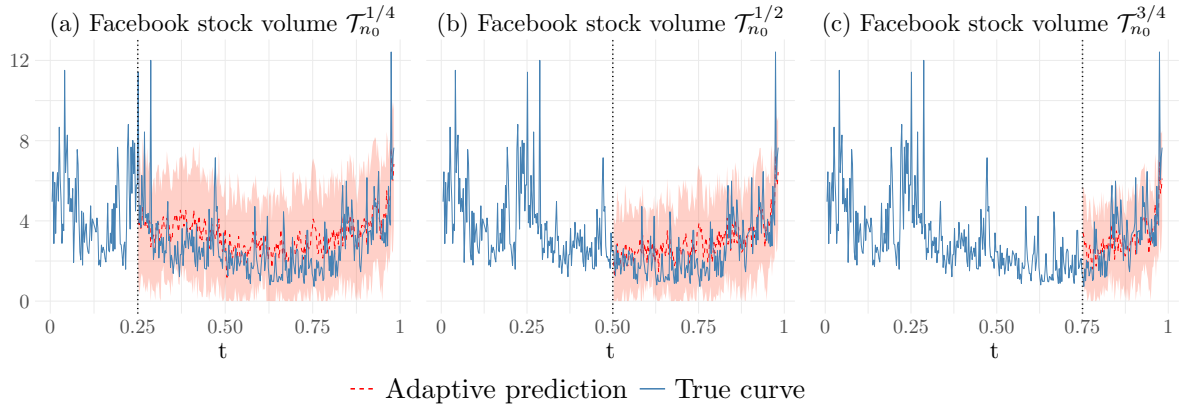


Figure 14: Predictions of the last Facebook stock volume curve under scenarios where observations are available in $(0, 1/4]$, $(0, 1/2]$, and $(0, 3/4]$, respectively. The solid line shows the true curve, the dashed line shows the predicted curve, and the shaded band indicates the 95% pointwise confidence interval. Stock volume values are normalised by 10^4 .

A Proofs associated to the Regularized linear predictor

Proof of Proposition 1. For simplicity, we write $\mathfrak{X}_{n_0+1}^{(M_{n_0})}(t)$ instead of $\mathfrak{X}_{n_0+1}^{(M_{n_0})}(t; \alpha, \sigma^2 \mathcal{D}_{n_0})$. We define the $(M_{n_0} \times M_{n_0})$ -matrices

$$\mathcal{A}_{M_{n_0}} = \alpha I_{M_{n_0}} + \sigma^2 \mathcal{D}_{n_0}, \quad R_{n_0, M_{n_0}} = (\Pi_{n_0, 0} \Pi_{n_0, 0}^* + \mathcal{A}_{M_{n_0}})^{-1},$$

and let $f = X_{n_0} - \mu$ hereafter. Notice that $R_{n_0, M_{n_0}}$ is symmetric with positive eigenvalues, and the largest one, denoted by $\lambda_{\max}(R_{n_0, M_{n_0}})$ is bounded by $1/\alpha$. We rewrite the Tikhonov regularized predictor as the sum of two terms,

$$\mathfrak{X}_{n_0+1, \alpha}^{(M_{n_0})}(s) = \mu(s) + \left\{ \Pi_{n_0, 1}^* R_{n_0, M_{n_0}} (\mathcal{D}_{n_0}^{1/2} \mathbf{Y}_{n_0} - \Pi_{n_0, 0} \mu) \right\}(s) = A_{n_0, \alpha}(s; f) + B_{n_0, \alpha}(s), \quad s \in I,$$

where

$$A_{n_0, \alpha}(s; f) = \mu(s) + \left[\Pi_{n_0, 1}^* \left\{ R_{n_0, M_{n_0}} \Pi_{n_0, 0} (X_{n_0} - \mu) \right\} \right](s) = C_{n_0, 1}(t)^\top \mathcal{D}_{n_0}^{1/2} R_{n_0, M_{n_0}} \Pi_{n_0, 0} (X_{n_0} - \mu),$$

and

$$B_{n_0, \alpha}(s) = \left[\Pi_{n_0, 1}^* \left\{ R_{n_0, M_{n_0}} \epsilon_{n_0} \right\} \right](s) = C_{n_0, 1}(s)^\top \mathcal{D}_{n_0}^{1/2} R_{n_0, M_{n_0}} \mathcal{D}_{n_0}^{1/2} \epsilon_{n_0}.$$

First, we derive a bound for $\mathbb{E}_{M, T}(\|B_{n_0}\|_\infty)$. Since (H4), we get,

$$\begin{aligned} \mathbb{E}_{M, T}(\|B_{n_0}\|_\infty^2) &= \sigma^2 \left\| C_{n_0, 1}(\cdot)^\top \mathcal{D}_{n_0}^{1/2} R_{n_0, M_{n_0}} \mathcal{D}_{n_0} R_{n_0, M_{n_0}} \mathcal{D}_{n_0}^{1/2} C_{n_0, 1}(\cdot) \right\|_\infty \\ &= \sigma^2 \left\| \text{Tr} \left(R_{n_0, M_{n_0}} \mathcal{D}_{n_0} R_{n_0, M_{n_0}} \mathcal{D}_{n_0}^{1/2} C_{n_0, 1}(\cdot) C_{n_0, 1}(\cdot)^\top \mathcal{D}_{n_0}^{1/2} \right) \right\|_\infty \\ &\leq \sigma^2 \lambda_{\max}^2(R_{n_0, M_{n_0}} \mathcal{D}_{n_0} R_{n_0, M_{n_0}}) \left\| \text{Tr} \left(\mathcal{D}_{n_0}^{1/2} C_{n_0, 1}(s) C_{n_0, 1}(s)^\top \mathcal{D}_{n_0}^{1/2} \right) \right\|_\infty \\ &\leq \frac{\sigma^2 \max_i \varrho_{n_0, i}}{\alpha^2} \left\| \sum_{j=1}^n \varrho_{n, j} C_1(s, T_{n_0, j})^2 \right\|_\infty \\ &\leq \sigma^2 \|c_1\|_\infty^2 \frac{\max_i \varrho_{n_0, i}}{\alpha^2}. \end{aligned} \tag{9}$$

For the second inequality in (9) we use the fact that

$$\lambda_{\max}^2(R_{n_0, M_{n_0}} \mathcal{D}_{n_0} R_{n_0, M_{n_0}}) \leq \max_i \varrho_{n_0, i} \lambda_{\max}^2(R_{n_0, M_{n_0}})^2 \leq \max_i \varrho_{n_0, i} / \alpha^2.$$

Indeed, for any unit vector $u \in \mathbb{R}^{M_{n_0}}$, $u^\top R_{n_0, M_{n_0}} \mathcal{D}_{n_0} R_{n_0, M_{n_0}} u = v^\top \mathcal{D}_{n_0} v$, where $v = R_{n_0, M_{n_0}} u$. Since $\mathcal{D}_{n_0}$ is a diagonal matrix with diagonal terms in $(0, 1)$,

$$u^\top R_{n_0, M_{n_0}} \mathcal{D}_{n_0} R_{n_0, M_{n_0}} u = \sum_{i=1}^n \varrho_{n_0, i} v_i^2 \leq \max_i \varrho_{n_0, i} \|R_{n_0, M_{n_0}} u\|_2^2 \leq \max_i \varrho_{n_0, i} \lambda_{\max}^2(R_{n_0, M_{n_0}})^2 \|u\|_2^2.$$

The bound on $\lambda_{\max}^2$ follows. As a consequence of (9), the rate of convergence to zero for $\mathbb{E}_{M, T}(\|B_{n_0}\|_{\mathbb{L}^2(I)}^2)$ is bounded by $\alpha^{-2} \max_i \varrho_{n_0, i}$. In the case of common design, $\max_i \varrho_{n_0, i} = M_{n_0}^{-1}$. **With independent design,** $\varrho_{n_0, i} = \{M_{n_0} \widehat{g}_{n_0, i}(T_{n_0, i})\}^{-1}$, it holds

$$\max_{1 \leq i \leq M_{n_0}} \varrho_{n_0, i} = \frac{1}{M_{n_0}} \cdot \max_{1 \leq i \leq M_{n_0}} \frac{1}{\widehat{g}_{n_0, i}(T_{n_0, i})}.$$

By Giné and Guillou (2002), the leave-one-out Parzen–Rosenblatt estimator satisfies

$$\sup_{1 \leq i \leq M_{n_0}} |\{g(T_{n_0, i}) / \widehat{g}_{n_0, i}(T_{n_0, i})\} - 1| \rightarrow 0, \quad \text{almost surely.}$$

Since $g_{\min} := \inf_{t \in I} g(t) > 0$ by assumption (H3), this uniform convergence implies that, almost surely for M_{n_0} large enough, $\widehat{g}_{n_0,i}(T_{n_0,i}) \geq g_{\min}/2$ for all $1 \leq i \leq M_{n_0}$. Consequently, $\max_{1 \leq i \leq M_{n_0}} \varrho_{n_0,i} \leq \frac{2}{M_{n_0} g_{\min}}$, which is of order $\mathcal{O}_{\mathbb{P}}(M_{n_0}^{-1})$, matching the common design case. We conclude that in both cases, We deduce that

$$\mathbb{E}_{M,T}(\|B_{n_0}\|_{\mathbb{L}^2(I)}^2) = \alpha^{-2} \mathcal{O}_{\mathbb{P}}(M_{n_0}^{-1}).$$

To study A_{n_0} , let $\beta_{n_0} = (\beta_{n_0,i}) = R_{n_0,M_{n_0}} \Pi_{n_0,0} f \in \mathbb{R}^{M_{n_0}}$ so that

$$A_{n_0}(s; f) - \mu(s) = \{\Pi_{n_0,1}^* \beta_{n_0}\}(s).$$

By definition, the vector β_{n_0} is then the unique solution of

$$(\Pi_{n_0,0} \Pi_{n_0,0}^* + \mathcal{A}_{M_{n_0}}) \beta_{n_0} = \mathcal{D}_{n_0}^{1/2} (f(T_{n_0,1}), \dots, f(T_{n_0,M_{n_0}}))^{\top}.$$

We prove below that with the continuous random function $\mathfrak{h} = (C_0 + \alpha \text{Id})^{-1} f$ we get

$$\|\beta_{n_0} - \Pi_{n_0,0} \mathfrak{h}\|_2 \rightarrow 0 \quad \text{in probability.} \quad (10)$$

For any $1 \leq i \leq M_{n_0}$, let

$$\delta_{n_0,i} = \beta_{n_0,i} - \sqrt{\varrho_{n_0,i}} \mathfrak{h}(T_{n_0,i}) \quad \text{and} \quad \delta_{n_0} = (\delta_{n_0,i}) \in \mathbb{R}^{M_{n_0}}.$$

such that $\delta_{n_0} = \beta_{n_0} - \Pi_{n_0,0} \mathfrak{h}$. Recall that, by definition, the function $\mathfrak{h}$ satisfies,

$$\int_0^1 \mathfrak{h}(s) c_0(s, T_{n_0,i}) ds + \alpha \mathfrak{h}(T_{n_0,i}) = f(T_{n_0,i}), \quad i = 1, \dots, M_{n_0}.$$

Let $\mathbf{r}_{n_0} = (r_{n_0,i}) := R_{n_0,M_{n_0}}^{-1} \delta_{n_0}$. Then we have

$$\mathbf{r}_{n_0} = \Pi_{n_0,0} f - \Pi_{n_0,0} [\Pi_{n_0,0}^* \Pi_{n_0,0} \mathfrak{h}] - \alpha \Pi_{n_0,0} \mathfrak{h} - \sigma^2 \mathcal{D}_{n_0} \Pi_{n_0,0} \mathfrak{h},$$

i.e. for $i = 1, \dots, M_{n_0}$,

$$r_{n_0,i} = \sqrt{\varrho_{n_0,i}} \int_0^1 \mathfrak{h}(s) c_0(s, T_{n_0,i}) ds - \sqrt{\varrho_{n_0,i}} \int_0^1 \mathfrak{h}(s) c_0(T_{n_0,i}, s) dQ_{n_0}(s) - \sigma^2 \varrho_{n_0,i} \sqrt{\varrho_{n_0,i}} \mathfrak{h}(T_{n_0,i}).$$

So we get that,

$$\begin{aligned} \|\mathbf{r}_{n_0}\|_2^2 &\leq 2 \sum_{i=1}^{M_{n_0}} \varrho_{n_0,i} \left(\int_0^1 \mathfrak{h}(s) c_0(s, T_{n_0,i}) ds - \int_0^1 \mathfrak{h}(s) c_0(T_{n_0,i}, s) dQ_{n_0}(s) \right)^2 + 2\sigma^2 \sum_{i=1}^{M_{n_0}} \varrho_{n_0,i}^3 \mathfrak{h}^2(T_{n_0,i}) \\ &\leq 2 \sum_{i=1}^{M_{n_0}} \varrho_{n_0,i} \left(\int_0^1 \mathfrak{h}(s) c_0(s, T_{n_0,i}) ds - \int_0^1 \mathfrak{h}(s) c_0(T_{n_0,i}, s) dQ_{n_0}(s) \right)^2 + 2\sigma^2 \{\|\mathfrak{h}\|_{\infty} \max_i \varrho_{n_0,i}\}^2 \\ &=: 2 \sum_{i=1}^{M_{n_0}} \varrho_{n_0,i} \left(\tilde{I}(\varphi_i) - I(\varphi_i) \right)^2 + 2\sigma^2 \{\|\mathfrak{h}\|_{\infty} \max_i \varrho_{n_0,i}\}^2, \end{aligned}$$

where $\varphi_i(\cdot) = \mathfrak{h}(\cdot) c_0(\cdot, T_{n_0,i})$ and, for any $\varphi : I \rightarrow \mathbb{R}$,

$$\tilde{I}(\varphi) = \int_0^1 \varphi(s) dQ_{n_0}(s) \quad \text{and} \quad I(\varphi) = \int_0^1 \varphi(s) ds.$$

The cases common and independent design have to be treated separately.

Independent design. Using the decomposition of $\tilde{I}(\varphi) - I(\varphi)$ as on [Delyon and Portier \(2016, p. 2196\)](#), we can write it as

$$\tilde{I}(\varphi_i) - I(\varphi_i) = R_{0,i} + R_{1,i} + R_{2,i}^{(1)} + R_{2,i}^{(2)} + R_{5,i}, \quad (11)$$

where $R_{1,i}$, $R_{2,i}^{(1)}$ and $R_{2,i}^{(2)}$ are the bias terms (with $R_{2,i}^{(1)} + R_{2,i}^{(2)}$ corresponding to R_2 in the decomposition used by [Delyon and Portier \(2016, Step 3, p. 2191\)](#)). In our case, let $R_{2,i}^{(2)}$ be the term with the rate depending on the smoothness of φ_i , while the rates of $R_{1,i}$ and $R_{2,i}^{(1)}$ depend on the smoothness of the density g . Following the lines of the proof of [Delyon and Portier \(2016, Theorem 1-\(i\)\)](#), the term $|M_{n_0}^{1/2} R_{2,i}^{(2)}|$ can be bounded by a product between a constant depending only on g and the kernel K , the Hölder constant of φ_i and h^β , see [Delyon and Portier \(2016, Eq. \(20\) in Lemma 6\)](#) and the decomposition in the last display before *Step 4* in [Delyon and Portier \(2016, on p. 2192\)](#). La régularité est $\min(\beta, r)$? Meanwhile, the sum $|R_{1,i} + R_{2,i}^{(1)}|$ can be written under the form of a product between $(\|\varphi_i\|_\infty + 1)$, the uniform bound of $g^{(r)}$ times h^r , and a constant depending only on g and K , see [Delyon and Portier \(2016, Eq. \(18\) and \(19\) in Lemma 6\)](#) and the bounds on R_2 in [Delyon and Portier \(2016, p. 2192\)](#). Moreover, the Hölder constant of φ_i is bounded by $\alpha^{-1}\|c_0\|_\infty \mathfrak{L}_{n_0}(\beta) + \alpha^{-1}\|f\|_\infty \sqrt{\|c_0\|_\infty} \sqrt{\mathbb{E} \mathfrak{L}_{n_0}(\beta)^2}$. Then, since both $\mathfrak{L}_{n_0}(\beta)$ and $\|X_{n_0} - \mu\|_\infty$ are independent of the design and have finite expectation, integrating with respect X_{n_0} , we deduce

$$\mathbb{E}_{M,T} \left[\sum_{i=1}^{M_{n_0}} \varrho_{n_0,i} \left(R_{1,i} + R_{2,i}^{(1)} + R_{2,i}^{(2)} \right)^2 \right] = \alpha^{-2} O_{\mathbb{P}}(h^{2r} + M_{n_0}^{-1} h^{2\beta}). \quad (12)$$

To bound the last term $\mathcal{V}_{M_{n_0}}(\varphi_i) = R_0 + R_5$ in the decomposition (11), we first note that by the definition of the terms in the decomposition,

$$\forall i, \quad |\mathcal{V}_{M_{n_0}}(\varphi_i)| \leq \alpha^{-1} \|f\|_\infty \sup_{\phi \in \mathcal{S}(\mathfrak{h})} |\mathcal{V}_{M_{n_0}}(\phi)|,$$

where

$$\mathcal{S}(\mathfrak{h}) = \{\phi = \alpha \mathfrak{h} \|f\|_\infty^{-1} c_0(\cdot, t), t \in I\}.$$

Note that $\|\phi\|_\infty \leq \|c_0\|_\infty, \forall \phi \in \mathcal{S}(\mathfrak{h})$. [Delyon and Portier \(2016\)](#) derive the rate $O_{\mathbb{P}}(M_{n_0}^{-1} h^{-1})$ for $\mathcal{V}_{M_{n_0}}(\phi)$ with a single ϕ . In a closely related context, [Cléménçon and Portier \(2018\)](#) extended the result and derived the bound $O_{\mathbb{P}}(\sqrt{M_{n_0}^{-1} h^{-1} \log(1/h)})$ for $\mathcal{V}_{M_{n_0}}(\cdot)$ uniformly over a countable VC class. The arguments of [Cléménçon and Portier \(2018\)](#), based on exponential bounds (depending only on the complexity and the envelope of the class, and not on the class itself) for deviations of empirical processes and degenerate U -processes, can also be applied in our case. Taking $\mathcal{S}_{\mathbb{Q}}(\mathfrak{h}) = \{\alpha \mathfrak{h} \|f\|_\infty^{-1} c_0(\cdot, t), t \in \mathbb{Q} \cap (0, 1)\}$ as a countable subset of $\mathcal{S}(\mathfrak{h})$, $\mathcal{S}(\mathfrak{h})$ is countably approximable in the sense of [Major \(2006\)](#). Indeed $t \mapsto c_0(s, t)$ is continuous uniformly in s , the map $t \mapsto \mathcal{V}_{M_{n_0}}(\phi_t)$ is p.s. continuous, so that the supremum over $t \in I$ equals p.s. the supremum over $t \in \mathbb{Q} \cap (0, 1)$. Moreover, by the uniform Hölder continuity of $c_0(s, \cdot)$ (uniformly in s), the class $\mathcal{S}_{\mathbb{Q}}(\mathfrak{h})$ is of VC-type with complexity constants independent of M_{n_0} and X_{n_0} (see Lemma 1). Applying the nonasymptotic bound of [Cléménçon and Portier \(2018, Lemma 3\)](#), which extends the concentration inequality of Major (2006) to degenerate U -processes indexed by VC-type classes, we obtain

$$\sup_{\phi \in \mathcal{S}(\mathfrak{h})} |\mathcal{V}_{M_{n_0}}(\phi)| = O_{\mathbb{P}} \left(\sqrt{M_{n_0}^{-1} h^{-1} \log(1/h)} \right)$$

Since $\|X_{n_0} - \mu\|_\infty$ is integrable and the exponential deviation bound for the supremum above depends on X_{n_0} only through $\|f\|_\infty$ (via the envelope of the class), we may integrate with respect to X_{n_0} to deduce

$$\mathbb{E}_{M,T} \left[\sum_{i=1}^{M_{n_0}} \varrho_{n_0,i} \mathcal{V}_{M_{n_0}}(\varphi_i)^2 \right] \leq \alpha^{-2} O_{\mathbb{P}}(\{M_{n_0}^{-1} h^{-1} \log(1/h)\}^2). \quad (13)$$

Common design. In this case we simply have to bound the error of a standard Riemann sum when the integrand is Hölder continuous. The details are omitted.

Now, using the bound on the spectral norm of $R_{n_0, M_{n_0}}$, we get

$$\|\delta_{n_0}\|_2^2 = \|\beta_{n_0} - \Pi_{n_0,0}h\|_2^2 \leq \alpha^{-2} \|\mathbf{r}_{n_0}\|_2^2.$$

Finally, by Cauchy-Schwarz inequality, for any $s \in I$ we get

$$\begin{aligned} |\{A_{n_0}(s; f) - \mu(s)\} - \Pi_{n_0,1}^* \Pi_{n_0,0} h(s)| &= |C_{1,n}(s)^\top \mathcal{D}_{n_0}^{1/2} (\beta_{n_0} - \Pi_{n_0,0} h)| \\ &\leq \|\mathcal{D}_{n_0}^{1/2} C_{1,n}(s)\|_2 \|\beta_{n_0} - \Pi_{n_0,0} h\|_2, \\ &\leq \|c_1\|_\infty \|\beta_{n_0} - \Pi_{n_0,0} h\|_2, \end{aligned} \quad (14)$$

and the last bound does not depend on s . Combining (12), (13) and (14), we get (10). $\square$

Lemma 1. Let $\Theta \subset \mathbb{R}^d$ be compact and let $\mathcal{F} = \{f_\theta : \theta \in \Theta\}$ be a class of uniformly bounded functions on $[0, 1]$ satisfying, for some $L > 0$ and $\alpha \in (0, 1]$,

$$\|f_{\theta_1} - f_{\theta_2}\|_\infty \leq L |\theta_1 - \theta_2|^\alpha, \quad \theta_1, \theta_2 \in \Theta.$$

Let $\mathbf{g} : [0, 1] \rightarrow \mathbb{R}$ be measurable with $\|\mathbf{g}\|_\infty \leq M$. Then, for all $\varepsilon > 0$ and all probability measures Q on $[0, 1]$,

$$N(\varepsilon, \mathbf{g}\mathcal{F}, L^2(Q)) \leq \left(\frac{c_\Theta ML}{\varepsilon} \right)^{d/\alpha}, \quad (15)$$

where $c_\Theta := (2 \text{diam}(\Theta))^\alpha$ depends only on α and Θ . In particular, $\mathbf{g}\mathcal{F}$ is a VC class with metric entropy exponent d/α , equal to that of $\mathcal{F}$.

Proof of Lemma 1. Set $\delta := (\varepsilon/(ML))^{1/\alpha}$. A standard volumetric argument yields a δ -net $\{\theta_1, \dots, \theta_N\} \subset \Theta$ with

$$N \leq \left(\frac{2 \text{diam}(\Theta)}{\delta} \right)^d = \left(\frac{c_\Theta ML}{\varepsilon} \right)^{d/\alpha}.$$

For any $\theta \in \Theta$, pick θ_i with $|\theta - \theta_i| \leq \delta$. Then

$$\|gf_\theta - gf_{\theta_i}\|_\infty \leq M \|f_\theta - f_{\theta_i}\|_\infty \leq ML \delta^\alpha = \varepsilon.$$

Hence $\{\mathbf{g}f_{\theta_1}, \dots, \mathbf{g}f_{\theta_N}\}$ is an ε -cover of $\mathbf{g}\mathcal{F}$ in L^∞ , and (15) follows since $\|\cdot\|_{L^2(Q)} \leq \|\cdot\|_\infty$ for any probability measure Q . $\square$

Proof of Proposition 2. We can decompose $\hat{\mathfrak{X}}_{n_0} - \mathfrak{X}_{n_0}$ as the sum of four terms,

$$\hat{\mathfrak{X}}_{n_0} - \mathfrak{X}_{n_0} = A + B + C + D$$

where

$$\begin{aligned} A &= \hat{\mu} - \mu, & B &= \hat{\Pi}_{n_0,1}^* \hat{R}_{n_0}^{-1} \Pi_{n_0,0} (\hat{\mu} - \mu), \\ C &= (\hat{\Pi}_{n_0,1}^* - \Pi_{n_0,1}^*) \hat{R}_{n_0}^{-1} (\mathcal{D}_{n_0}^{1/2} \mathbf{Y}_{n_0} - \Pi_{n_0,0} \mu), & D &= \Pi_{n_0,1}^* (\hat{R}_{n_0}^{-1} - R_{n_0}^{-1}) (\mathcal{D}_{n_0}^{1/2} \mathbf{Y}_{n_0} - \Pi_{n_0,0} \mu). \end{aligned}$$

Remark that,

$$\|\Pi_{n_0,0}(\hat{\mu} - \mu)\|_2^2 = \sum_{i=1}^{M_n} \rho_{n,i} (\hat{\mu}(T_{n,i}) - \mu(T_{n,i}))^2 \leq \|A\|_\infty^2.$$

Since the largest eigenvalue of $\widehat{R}_{n_0, M_{n_0}}$ is bounded by $1/\alpha$, and according to Theorem 1 and 3, we have,

$$\|A + B\|_\infty = \|\widehat{\mu} - \mu\|_\infty (1 + \alpha^{-1}(\|c_1\|_\infty + \|\widehat{c}_1 - c_1\|_\infty)).$$

Furthermore, we can establish that,

$$\|\mathcal{D}_{n_0}^{1/2} \mathbf{Y}_{n_0} - \Pi_{n_0, 0} \mu\|_2 \leq \sqrt{2} \sqrt{\|X_{n_0} - \mu\|_\infty^2 + \sigma^2 \left(1 + \mathcal{O}_{\mathbb{P}} \left(\max_i \rho_{n_0, i}\right)\right)}$$

Indeed, using that $(a + b)^2 \leq 2a^2 + 2b^2$ for $a, b \in \mathbb{R}$, we have,

$$\|\mathcal{D}_{n_0}^{1/2} \mathbf{Y}_{n_0} - \Pi_{n_0, 0} \mu\|_2^2 \leq 2 \sum_{i=1}^{M_n} \rho_{n_0, i} (X_{n_0}(T_{n_0, i}) - \mu(T_{n_0, i}))^2 + 2\sigma^2 \sum_{i=1}^{M_n} \rho_{n_0, i} \epsilon_{n, i}^2.$$

Since $\rho_{n_0, 1} + \dots + \rho_{n_0, M_{n_0}} = 1$,

$$\begin{aligned} \sum_{i=1}^{M_n} \rho_{n_0, i} (X_{n_0}(T_{n_0, i}) - \mu(T_{n_0, i}))^2 &\leq \|X_{n_0} - \mu\|_\infty^2 \\ \mathbb{E}_{M, T} \left(\sum_{i=1}^{M_n} \rho_{n_0, i} \epsilon_{n, i}^2 \right) &= 1 \quad \text{and} \quad \text{Var}_{M, T} \left(\sum_{i=1}^{M_n} \rho_{n_0, i} \epsilon_{n, i}^2 \right) \leq \max_i \rho_{n_0, i}. \end{aligned}$$

By Tchebychev inequality, we deduce that,

$$\sigma^2 \sum_{i=1}^{M_n} \rho_{n_0, i} \epsilon_{n, i}^2 = \sigma^2 \left(1 + \mathcal{O}_{\mathbb{P}}(\max_i \rho_{n_0, i})\right).$$

Then the Cauchy-Schwarz inequality implies that,

$$\begin{aligned} |C(t)| &\leq \|\widehat{c}_1 - c_1\|_\infty \left\| \widehat{R}_{n_0}^{-1} (\mathcal{D}_{n_0}^{1/2} \mathbf{Y}_{n_0} - \Pi_{n_0, 0} \mu) \right\|_2 \\ &\leq \alpha^{-1} \|\widehat{c}_1 - c_1\|_\infty \sqrt{2\|X_{n_0} - \mu\|_\infty^2 + 2\sigma^2 \left(1 + \mathcal{O}_{\mathbb{P}} \left(\max_i \rho_{n_0, i}\right)\right)} \end{aligned}$$

Remark that D can be rewritten as,

$$D = \Pi_{n_0, 1}^* \widehat{R}_{n_0}^{-1} (R_{n_0} - \widehat{R}_{n_0}) R_{n_0}^{-1} (\mathcal{D}_{n_0}^{1/2} \mathbf{Y}_{n_0} - \Pi_{n_0, 0} \mu).$$

Using the same argument than the term C , we deduce that,

$$\begin{aligned} |D(t)| &\leq \|c_1\|_\infty \|\widehat{R}_{n_0}^{-1}\|_2 \|\Pi_{n_0, 0} (\widehat{\Pi}_{n_0, 0}^* - \Pi_{n_0, 0}^*) + (\widehat{\sigma} - \sigma^2) \mathcal{D}_{n_0}\|_2 \|R_{n_0}^{-1}\|_2 \|\mathcal{D}_{n_0}^{1/2} \mathbf{Y}_{n_0} - \Pi_{n_0, 0} \mu\|_2 \\ &\leq \|c_1\|_\infty \sqrt{2\|X_{n_0} - \mu\|_\infty^2 + 2\sigma^2 \left(1 + \mathcal{O}_{\mathbb{P}} \left(\max_i \rho_{n_0, i}\right)\right)} \left(\|\widehat{c}_0 - c_0\|_\infty + |\widehat{\sigma} - \sigma| \max_i \rho_{n, i} \right) \alpha^{-2}. \end{aligned}$$

□

B Uniform consistent estimators

Theorem 1. Assume the conditions (H1) to (H5) and (H??) to (H??) hold true. For any $\underline{H} \in [2/(p-4), \inf H_t]$, we have:

$$\sup_{t \in I} |\widehat{\mu}_N(t; h) - \mu(t)| = \mathcal{O}_{\mathbb{P}} \left(h^{\underline{H}} + \sqrt{\frac{\log(N\lambda)}{N \min(1, (\lambda h)^{2-2/p})}} + \sqrt{\frac{\log(N\lambda)}{N}} \right),$$

which is minimized by

$$\underline{h}^* = \mathcal{O}_{\mathbb{P}} \left(\max \left\{ \{\log(N\lambda)/(N\lambda^{2-2/p})\}^{1/\{2\underline{H}+2-2/p\}}, \{\log(N\lambda)/(N)\}^{1/\{2\underline{H}\}} \right\} \right),$$

and the estimator $\hat{\mu}_N^*(t) = \hat{\mu}_N(t; \underline{h}^*)$ of $\mu(t)$ satisfies

$$\sup_{t \in I} |\hat{\mu}_N^*(t) - \mu(t)| = \mathcal{O}_{\mathbb{P}} \left(\{\log(N\lambda)/(N\lambda^{2-2/p})\}^{\underline{H}/\{2\underline{H}+2-2/p\}} + \{\log(N\lambda)/N\}^{1/2} \right).$$

Theorem 2. Assume the conditions (H1) to (H5) and (H??) to (H??), hold true. Then, for any $\underline{H} \in [2/(p-4), \inf H_t)$, we have:

$$\begin{aligned} \sup_{s, t \in I} |\hat{c}_{N, \ell}(s, t; h_s, h_t) - c_{\ell}(s, t)| &= \mathcal{O}_{\mathbb{P}} \left\{ h_s^{\underline{H}} + h_t^{\underline{H}} + \sqrt{\log(N\lambda)/N} \right. \\ &\quad \left. + \sqrt{\log(N\lambda)/\{N \min(1, (\lambda h_s)^{2-2/p}) \min(1, (\lambda h_t)^{2-2/p})\}} \right\}, \end{aligned}$$

which is minimised by h_s^* and h_t^* which have the same rate,

$$\mathcal{O}_{\mathbb{P}} \left(\max \left\{ \{\log(N\lambda)/N\}^{1/\{2\underline{H}\}}, \{\log(N\lambda)/(N\lambda^{2-2/p})\}^{1/\{2\underline{H}+2-2/p\}}, \right. \right. \\ \left. \left. \{\log(N\lambda)/(N\lambda^{4-4/p})\}^{1/\{2\underline{H}+4-4/p\}} \right\} \right),$$

and the estimator $\hat{\Gamma}_{N, \ell}^*(s, t) = \hat{\Gamma}_{N, \ell}(s, t; h_s^*, h_t^*)$ of $\Gamma_{\ell}(s, t)$ satisfies

$$\begin{aligned} \sup_{s, t \in I} |\hat{c}_{N, \ell}^*(s, t) - c_{\ell}(s, t)| &= \mathcal{O}_{\mathbb{P}} \left(\{\log(N\lambda)/(N\lambda^{2-2/p})\}^{\underline{H}/\{2\underline{H}+2-2/p\}} \right. \\ &\quad \left. + \{\log(N\lambda)/(N\lambda^{4-4/p})\}^{\underline{H}/\{2\underline{H}+4-4/p\}} + \sqrt{\log(N\lambda)/N} \right). \end{aligned}$$

Theorem 3. Assume the conditions (H1) to (H5) and (H??) to (H??) hold true. Let $\ell = 0$. Then, for any $\underline{H} \in [2/(p-4), \inf H_t)$:

1. Outside the diagonal set, we have

$$\begin{aligned} \sup_{\substack{s, t \in I \\ |s-t| > 2 \max \mathcal{H}_N}} |\hat{c}_{N, 0}(s, t; h_s, h_t) - c_0(s, t)| &= \mathcal{O}_{\mathbb{P}} \left\{ h_s^{\underline{H}} + h_t^{\underline{H}} + \sqrt{\log(N\lambda)/N} \right. \\ &\quad \left. + \sqrt{\log(N\lambda)/\{N \min(1, (\lambda h_s)^{2-2/p}) \min(1, (\lambda h_t)^{2-2/p})\}} \right\}, \end{aligned}$$

which is minimised by h_s^* and h_t^* which have the same rate,

$$\mathcal{O}_{\mathbb{P}} \left(\max \left\{ \{\log(N\lambda)/N\}^{1/\{2\underline{H}\}}, \{\log(N\lambda)/(N\lambda^{2-2/p})\}^{1/\{2\underline{H}+2-2/p\}}, \right. \right. \\ \left. \left. \{\log(N\lambda)/(N\lambda^{4-4/p})\}^{1/\{2\underline{H}+4-4/p\}} \right\} \right),$$

and the estimator $\tilde{c}_{N, 0}^*(s, t) = \hat{c}_{N, 0}(s, t; h_s^*, h_t^*)$ of $\Gamma_0(s, t)$ satisfies

$$\begin{aligned} \sup_{\substack{s, t \in I \\ |s-t| > 2 \max \mathcal{H}_N}} |\tilde{c}_{N, 0}^*(s, t) - c_0(s, t)| &= \mathcal{O}_{\mathbb{P}} \left(\{\log(N\lambda)/(N\lambda^{2-2/p})\}^{\underline{H}/\{2\underline{H}+2-2/p\}} \right. \\ &\quad \left. + \{\log(N\lambda)/(N\lambda^{4-4/p})\}^{\underline{H}/\{2\underline{H}+4-4/p\}} + \sqrt{\log(N\lambda)/N} \right). \end{aligned}$$

2. On the diagonal line $s = t$, we have

$$\sup_{s \in I} |\widehat{c}_{N,0}(s, s; h_s, h_s) - c_0(s, s)| = \mathcal{O}_{\mathbb{P}} \left\{ h_s^{\frac{H}{2}} + \sqrt{\log(N\lambda) / \{N \min(1, \lambda h_s)^{2-2/p}\}} + \sqrt{\log(N\lambda) / N} \right\},$$

which is minimised by

$$h_s^* = \mathcal{O}_{\mathbb{P}} \left(\max \left\{ \{\log(N\lambda) / (N\lambda^{2-2/p})\}^{1/\{2H+2-2/p\}}, \{\log(N\lambda) / (N)\}^{1/\{2H\}} \right\} \right),$$

and the estimator $\widehat{c}_{N,0}^*(s, s) = \widehat{c}_{N,0}(s, s; h_s^*, h_s^*)$ of $\Gamma_0(s, s)$ satisfies

$$\sup_{s \in I} |\widehat{c}_{N,0}^*(s, s) - c_0(s, s)| = \mathcal{O}_{\mathbb{P}} \left(\{\log(N\lambda) / (N\lambda^{2-2/p})\}^{\frac{H}{\{2H+2-2/p\}}} + \{\log(N\lambda) / N\}^{1/2} \right).$$

3. On the diagonal band, for any $(s, t) \in I^2$ such that $|s - t| \leq 2 \max \mathcal{H}_N$, we have

$$\begin{aligned} \sup_{\substack{s, t \in I \\ |s-t| \leq 2 \max \mathcal{H}_N}} |\widehat{c}_{N,0}^*(s, t) - c_0(s, t)| &= \mathcal{O}_{\mathbb{P}} \left(\{\log(N\lambda) / (N\lambda^{2-2/p})\}^{\frac{H}{\{2H+2-2/p\}}} \right. \\ &\quad \left. + \{\log(N\lambda) / (N\lambda^{4-4/p})\}^{\frac{H}{\{2H+4-4/p\}}} + \sqrt{\log(N\lambda) / N} \right). \end{aligned}$$

References

- Bosq, D. (2000). *Linear processes in function spaces*, volume 149 of *Lecture Notes in Statistics*. Springer-Verlag, New York. Theory and applications.
- Cl  men  on, S. and Portier, F. (2018). Beating Monte Carlo integration: a nonasymptotic study of kernel smoothing methods. In Storkey, A. and Perez-Cruz, F., editors, *Proceedings of the Twenty-First International Conference on Artificial Intelligence and Statistics*, volume 84 of *Proceedings of Machine Learning Research*, pages 548–556. PMLR.
- Delyon, B. and Portier, F. (2016). Integral approximation by kernel smoothing. *Bernoulli*, 22(4):2177–2208.
- Gin  , E. and Guillou, A. (2002). Rates of strong uniform consistency for multivariate kernel density estimators. *Annales de l’Institut Henri Poincar   (B) Probability and Statistics*, 38(6):907–921.
- H  rmann, S. and Kokoszka, P. (2012). Functional time series. In *Handbook of statistics*, volume 30, pages 157–186. Elsevier.
- Horv  th, L., Rice, G., and Whipple, S. (2016). Adaptive bandwidth selection in the long run covariance estimator of functional time series. *Computational Statistics & Data Analysis*, 100:676–693.
- K  hnert, S., Rice, G., and Aue, A. (2026). Estimating invertible processes in Hilbert spaces, with applications to functional ARMA processes. *Bernoulli*, 32(2):1523 – 1546.
- Maissoro, H., Patilea, V., and Vimond, M. (2025). Adaptive estimation for weakly dependent functional times series. *Journal of Time Series Analysis*.
- Major, P. (2006). An estimate on the supremum of a nice class of stochastic integrals and u-statistics. *Probability Theory and Related Fields*, 134(3):489–537.
- Open Data R  seaux   nergies, O. (2024). Donn  es   co2mix nationales temps r  el. Accessed: 2024-11-04.
- Paparoditis, E. and Shang, H. L. (2023). Bootstrap prediction bands for functional time series [Correction]. *J. Amer. Statist. Assoc.*, 118(543):2212.

- Robinson, G. K. (1991). That BLUP is a good thing: the estimation of random effects. *Statist. Sci.*, 6(1):15–51. With comments and a rejoinder by the author.
- Rubín, T. and Panaretos, V. M. (2020). Sparsely observed functional time series: estimation and prediction. *Electron. J. Stat.*, 14(1):1137–1210.
- Stoev, S. A. and Taqqu, M. S. (2006). How rich is the class of multifractional Brownian motions? *Stochastic Processes and their Applications*, 116(2):200–221.
- Zhao, Y. (2026). Functional autoregression without truncation: A continuous-regularization approach. *arXiv preprint arXiv:2604.25205*.